

SUMMER INTERNSHIP REPORT

ON

“A Physics-Guided Health Indicator Framework Using Vibration Analysis for Material State Identification and Transition Monitoring in Drilling”

Submitted

by

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Abstract

Material transition detection during drilling is a critical challenge in manufacturing and Prognostics and Health Management, particularly for multi-layered structures where each layer has different physical properties. Most existing monitoring methods rely on empirical thresholds or machine learning models that are process-specific, require large training datasets, and offer limited physical explanation for their decisions.

This report proposes a Physics-Guided Health Indicator Framework that estimates the drilling condition directly from vibration measurements without using any machine learning or artificial intelligence. The framework combines structural dynamics, signal processing, information theory, and Kalman filter state estimation into a single methodology that is physically interpretable at every step.

The material's natural frequency, measured independently through Resonance Acoustic Pulse testing, serves as a physical reference connecting the free-vibration characteristics of each material to the vibration observed during drilling. Vibration features extracted using a sliding window approach are evaluated for repeatability, objectively weighted using the Entropy Weight Method, and fused into a Feature Fusion Index. A Kalman filter then estimates the latent Health Indicator representing the tool-material engagement state and its rate of change.

The framework is validated through repeated drilling experiments on four engineering materials. Repeatability, monotonicity, and physical consistency are investigated to confirm that the Health Indicator responds to real material physics rather than reflecting statistical processing alone. The results show that the proposed framework provides a material-independent, training-free foundation for drilling condition monitoring that can support future integration with machine learning, explainable AI, and closed-loop drilling control.

Keywords: Prognostics and Health Management; Drilling Monitoring; Physics-Guided Health Indicator; Kalman Filter; Entropy Weight Method; Structural Dynamics; Material Transition Detection.

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1. INTRODUCTION

1.1 Background and Problem

Modern manufacturing is moving toward autonomous, self-monitoring systems where the machine understands its own condition without human intervention. Drilling is one of the most common operations in this environment, used in aerospace, automotive, biomedical, and structural assembly, and it presents a particular monitoring challenge when the workpiece consists of multiple material layers stacked together.

When a drill passes through a layered structure, each layer behaves differently. The density, stiffness, damping, and internal structure of each material produce a different vibration response at the accelerometer. This means the vibration signal carries real physical information about what the drill is currently cutting through, when it is interpreted correctly.

The measured vibration can be expressed as

$$x(t) = f(M, T, P, S, N)$$

where M represents material properties, T is the tool condition, P is process parameters, S is the machine's structural dynamics, and N is measurement noise. This equation makes clear that the signal is not generated by one source. It is a mixture of everything happening simultaneously. The monitoring challenge is separating the material contribution from the rest.

$(\rho)(k)(c)(f_n)$ From a structural dynamics perspective, every material has characteristic properties, such as density, stiffness, damping, and natural frequency, which govern how it stores and releases vibration energy during cutting. A change in material, therefore, produces a measurable change in the vibration response. The question is whether that change can be tracked reliably in real time.

Within PHM, monitoring approaches fall into three broad categories: similarity-based models that compare signals to stored references, survival-based models that estimate remaining life from degradation statistics, and degradation-based models that continuously track the evolving system state. This work belongs to the third category.

The goal is not to compare signals to a library or predict when the tool will fail. The goal is to continuously estimate the physical state of the drilling process through a quantity called a Health Indicator, defined as

$$H(t) = g(x(t))$$

where $H(t)$ is the latent drilling condition and $g(\cdot)$ is the mapping from measured vibration to a physically meaningful state. An effective Health Indicator should evolve continuously with the drilling process, remain stable under measurement noise, carry physical meaning about the tool-material interaction, and provide a reliable basis for detecting when the drill is approaching a layer transition.

These requirements motivate the framework proposed in this paper.

1.2 Evolution of Drilling Condition Monitoring and Existing Research

Early threshold-based methods monitored cutting force, spindle current, torque, acoustic emission, and vibration directly. When a measured quantity crossed a preset threshold, an alarm was triggered. These methods are simple and still used in industry, but their thresholds are specific to one material, one tool, and one set of process parameters. Change any of these, and the threshold needs to be reset. In multi-layer drilling, this is impractical.

Signal processing and feature engineering improved on this by extracting descriptors from the raw signal rather than monitoring it directly. Time-domain features such as RMS, variance, kurtosis, crest factor, and skewness quantify energy, impulsiveness, and signal shape. Frequency-domain features such as spectral energy, centroid, entropy, and bandwidth describe how energy is distributed across frequencies. Time-frequency methods, including STFT, wavelet transforms, and EMD, capture transient events that neither domain alone can describe fully.

The limitation of feature-based approaches is that a single drilling signal can generate dozens of candidate features, many of which are redundant or sensitive to operating conditions rather than material properties. Selecting features that are genuinely repeatable and physically meaningful is not straightforward.

Machine learning addressed this by learning relationships between features and conditions from data rather than selecting them manually. SVM, random forests, neural networks, CNN, LSTM, and transformer models have all been applied to tool condition monitoring and material classification with impressive accuracy. However, their performance depends on the quality and quantity of labelled training data. When materials, tools, or conditions change, models trained under one configuration often fail under another. More importantly, the internal reasoning of these models is difficult to connect to the physical mechanics of drilling.

Explainable AI partially addresses this by showing which features influenced a model's decision. SHAP, LIME, and attention mechanisms make trained models more transparent. But they explain the model's behaviour. They do not validate whether the underlying Health Indicator itself represents real drilling physics. A model that explains a physically meaningless variable clearly is not more useful than one that does not.

This progression shows a consistent pattern. Each advance has improved computational performance while the physical interpretability of the monitored variable has received comparatively less attention.

1.3 Critical Research Gaps and Research Motivation

Despite the progress achieved in drilling condition monitoring, several challenges remain in the construction and interpretation of Health Indicators. Most existing studies focus on improving prediction accuracy through advanced computational models, while less attention is given to whether the monitored variable represents the physical evolution of drilling. As a result, the reliability of many intelligent monitoring systems is governed more by computational performance than by physical interpretability.

1.3.1 Observed Gap I: Prediction Accuracy versus Physical Representation

Modern drilling monitoring frameworks are commonly evaluated using statistical measures such as classification accuracy, prediction error, precision, recall, or F-score. Although these indicators effectively quantify computational performance, they do not necessarily demonstrate that the estimated Health Indicator represents the actual degradation behaviour occurring during drilling.

From an engineering perspective, a Health Indicator should evolve continuously with the interaction between the cutting tool and the workpiece. Therefore, improving prediction accuracy alone cannot guarantee that the monitored variable possesses physical significance. A physically meaningful degradation representation is required before intelligent decision-making can be considered reliable.

1.3.2 Observed Gap II: Feature Engineering versus Physics-Based Feature Selection

Vibration analysis provides a wide variety of candidate features derived from the time domain, frequency domain, and time-frequency domain. However, feature selection is frequently performed according to previous literature, engineering intuition, or machine learning performance. Such approaches do not necessarily ensure that the selected features remain stable and repeatable during repeated drilling experiments.

A physically interpretable Health Indicator should be constructed from features that exhibit high repeatability, sensitivity to process evolution, and consistency under identical operating conditions. Therefore, feature stability should be experimentally verified before feature fusion rather than being assumed a priori.

1.3.3 Observed Gap III: Statistical Validation versus Physical Validation

Another important limitation concerns the validation of Health Indicators. Existing studies generally evaluate Health Indicators using computational metrics, while comparatively little emphasis is placed on validating whether the estimated Health Indicator remains consistent with independently measurable physical properties of the material.

The dynamic response observed during drilling is governed by measurable material characteristics, including density, stiffness, damping, and structural natural frequency. Consequently, a physically meaningful Health Indicator should respond consistently to these material properties. Establishing such a relationship provides engineering confidence that the Health Indicator represents the actual tool-material interaction rather than simply reflecting mathematical processing of vibration signals.

1.3.4 Observed Gap IV: Artificial Intelligence before Physical Intelligence

Recent developments in Machine Learning, Deep Learning, Explainable Artificial Intelligence, and Digital Twins have significantly enhanced intelligent manufacturing systems. Nevertheless, the reliability of every intelligent algorithm fundamentally depends on the quality of the information supplied to it.

If the Health Indicator itself does not accurately represent the degradation behaviour of the drilling process, increasing the complexity of learning algorithms cannot guarantee physically reliable predictions. Therefore, a physically interpretable degradation model should first be established before introducing intelligent learning algorithms.

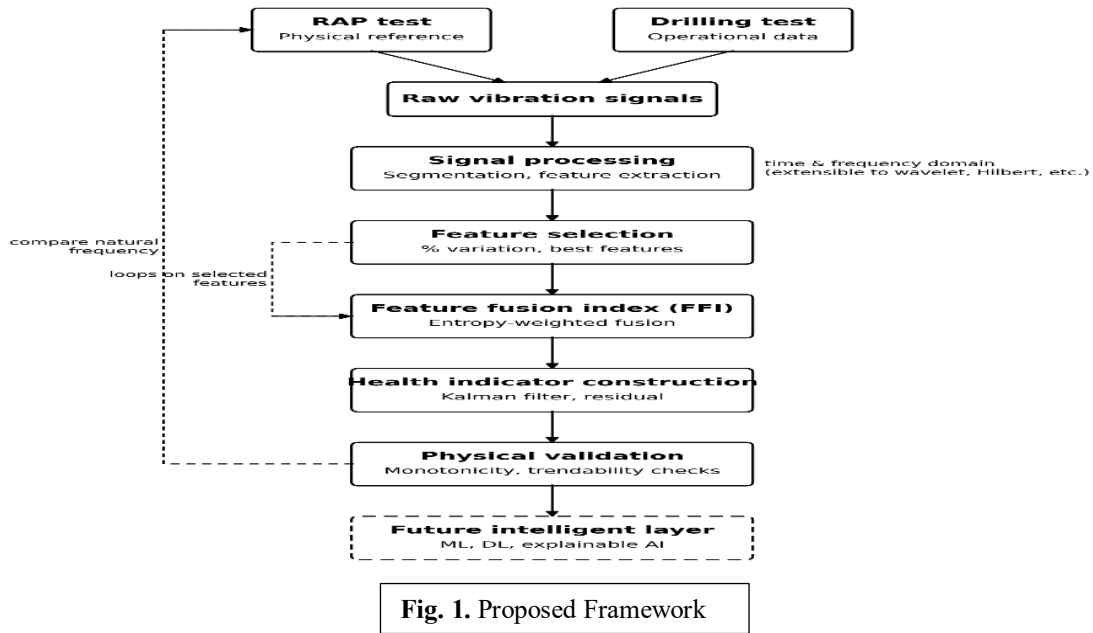
1.3.5 Research Motivation

Motivated by these research gaps, this work proposes a **Physics-Guided Health Indicator Framework** for drilling process monitoring. Unlike conventional approaches that directly employ extracted features for prediction or classification, the proposed framework first constructs a physically interpretable degradation representation of the drilling process using structural dynamics, vibration signal processing, information theory, and state estimation.

The philosophy adopted in this work is that **artificial intelligence should be built upon a physically validated Health Indicator rather than using machine learning to compensate for an unreliable physical representation**. Accordingly, vibration features are first evaluated for their physical stability and repeatability, followed by objective feature weighting, feature fusion, and latent state estimation to construct a smooth Health Indicator representing the evolving drilling condition.

Furthermore, the proposed methodology is designed as a generalized degradation modelling framework rather than a similarity-based diagnosis or survival prediction model. Its objective is to estimate the continuous degradation state of the drilling process and establish physically meaningful thresholds for identifying material engagement and transition. Once this physical foundation has been validated experimentally, methods such as Machine Learning, Deep Learning, Explainable Artificial Intelligence, Digital Twins, and uncertainty quantification can be incorporated to improve predictive capability without compromising physical interpretability.

2. Proposed Approach



2.1 Experimental Setup and Data Acquisition

To investigate vibration characteristics during drilling across different materials, an experimental setup was developed consisting of a drilling machine, an IEPE piezoelectric accelerometer, a Dewesoft data acquisition system, and a computer running DewesoftX software for real-time signal recording.

2.2 Materials

Four materials were selected to cover a wide range of densities and structural properties: wood, decorative panel (DD), acrylic, and thermocol. This selection ensures the proposed framework is tested under significantly different drilling conditions rather than being optimized for a single material type.

Specimen dimensions were measured using a digital Vernier calliper at three positions left, centre, and right, and the average was used to compute the dimensions and volume. Mass was measured using a digital weighing balance. Density was calculated as:

$$\rho = \frac{m}{V}$$

where m is mass in grams and V is volume in cm^3 . The physical properties of each specimen are summarized in

Material	Mass (g)	Length (mm)	Width (mm)	Thickness (mm)	Volume (cm ³)	Density (g/cm ³)	Density (kg/m ³)
Wood	148	261.175	79.1133	11.17	230.8266	0.6412	641.2
Decorative Panel	107	317.5	76.0266	3.2033	77.3155	1.3839	1383.9
Acrylic	11.0089	170.526	24.4433	2.36	9.8419	1.1186	1118.6
Thermocol	4.8621	141.57	125.8	22.33	397.6857	0.01223	12.23

Table 1.

The four materials span nearly two orders of magnitude in density, from the soft foam of thermocol to the dense decorative laminate panel. This range is intentional it tests whether the proposed Health Indicator framework can distinguish materials whose physical properties differ substantially rather than marginally.

2.3 Data Acquisition

Vibration signals were acquired using an IEPE accelerometer with a sensitivity of 100 mV/g, connected to the Dewesoft DAQ system. Signals were recorded in units of g at a sampling frequency of 20 kHz with a high-pass filter of 1.4 Hz, while the low-pass filter was disabled to preserve the full vibration response. The acquisition settings are shown in Table 2. The 20 kHz sampling frequency satisfies the Nyquist criterion for the frequency range of interest and captures both low-frequency structural responses and higher-frequency dynamic content generated during drilling. The resulting Nyquist frequency is 10 kHz, giving a usable analysis range of 0 to 10 kHz.

The same sensor configuration, mounting location, and acquisition parameters were maintained across all experiments to ensure consistency.

Parameter	Value
Sensor Type	IEPE Accelerometer
Measured Quantity	Acceleration
Unit	g
Sensor Sensitivity	100 mV/g
Sampling Frequency	20 kHz
Measurement Range	±10000 mV
High-pass Filter	1.4 Hz
Low-pass Filter	Off

Table 2.

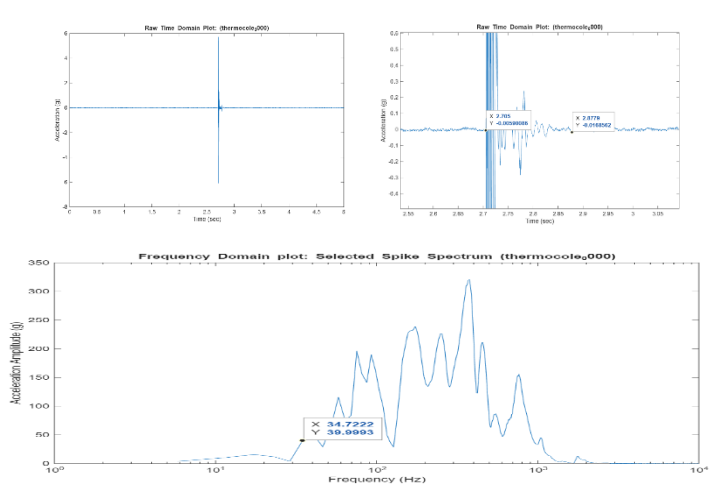
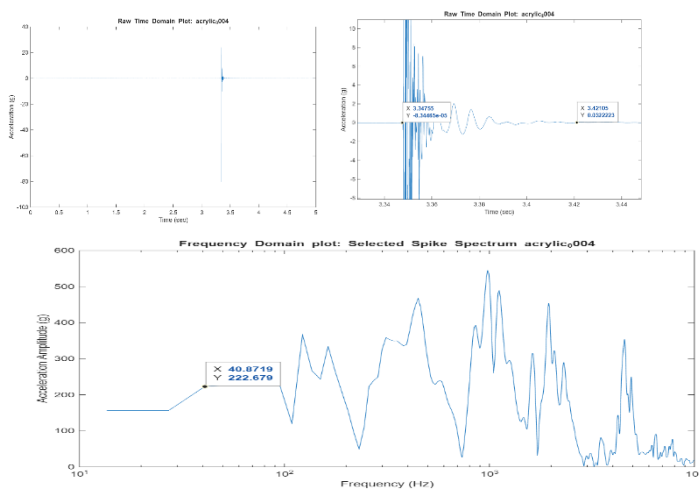
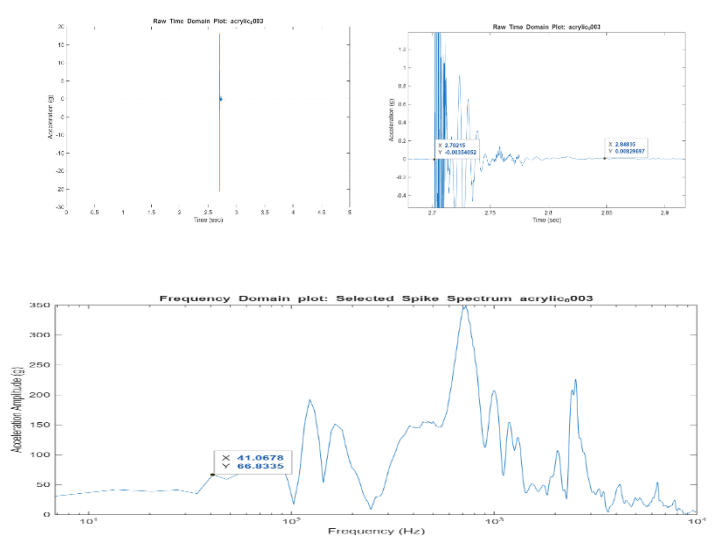
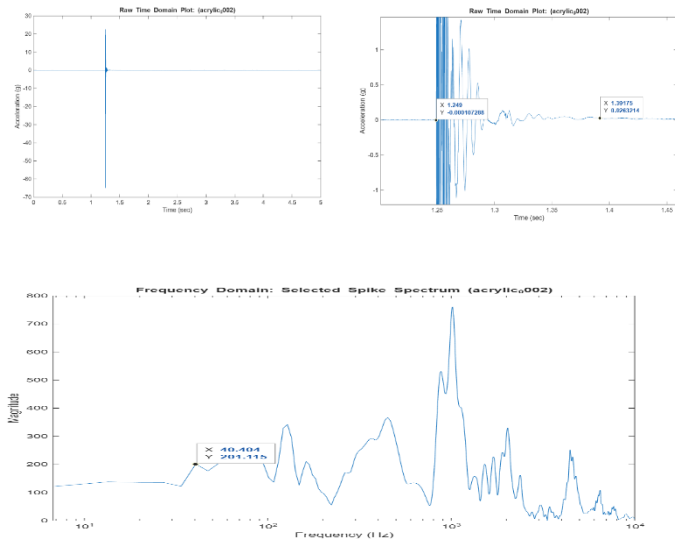
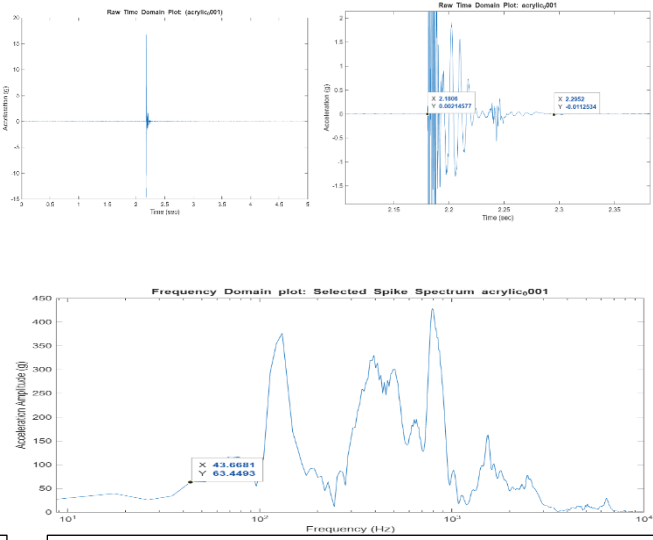
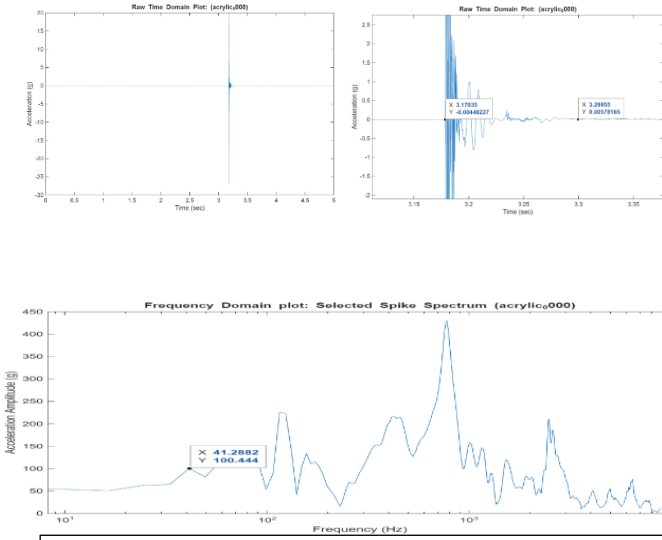
2.4 Resonance Acoustic Pulse Test

Before drilling, each material was characterized through a Resonance Acoustic Pulse (RAP) test to identify its dominant natural frequency. Each specimen was struck with a controlled impact while the accelerometer recorded the free vibration response. The test was repeated five times per material to assess repeatability.

The acquired time-domain response was transformed to the frequency domain using the Fast Fourier Transform:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

f_n The dominant peak in the resulting spectrum identifies the material's first natural frequency, which governs the frequency at which the material most readily stores and releases vibration energy. This natural frequency serves as a physics-based reference throughout the framework by connecting the free-vibration characteristics of each material to the vibration observed during active drilling.



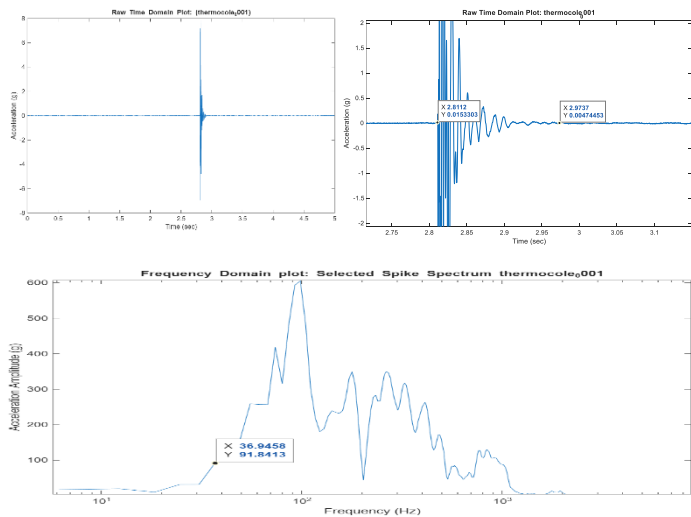


Fig. 8. Rap Test Time and frequency domain plots of Thermocole_1

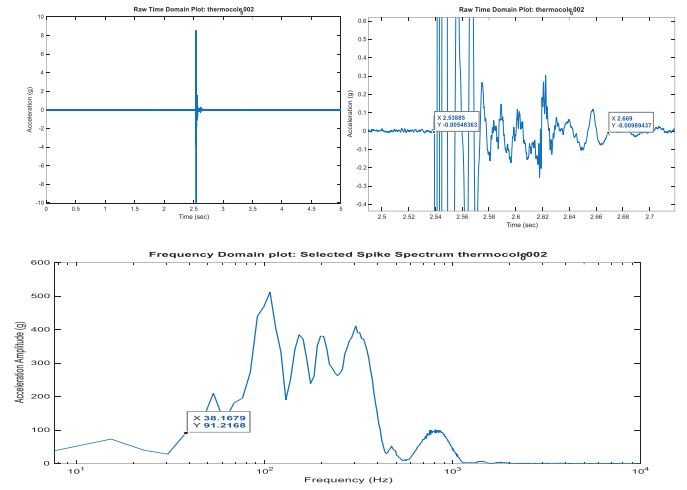


Fig. 9. Rap Test Time and frequency domain plots of Thermocole_2

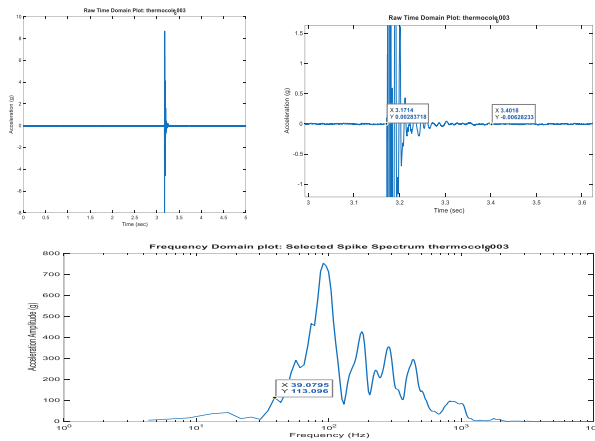


Fig. 10. Rap Test Time and frequency domain plots of Thermocole_3

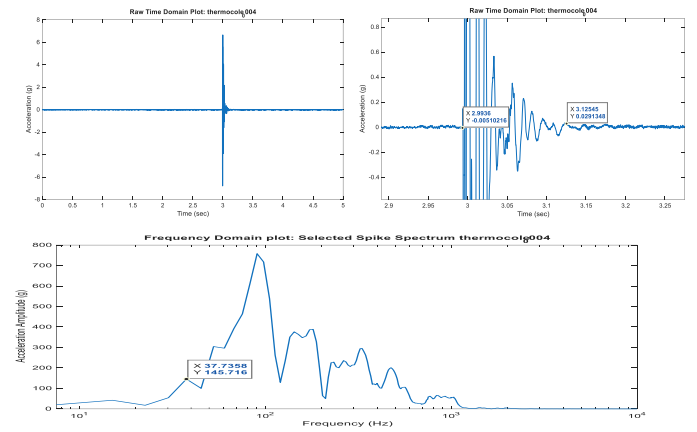


Fig. 11. Rap Test Time and frequency domain plots of Thermocole_4

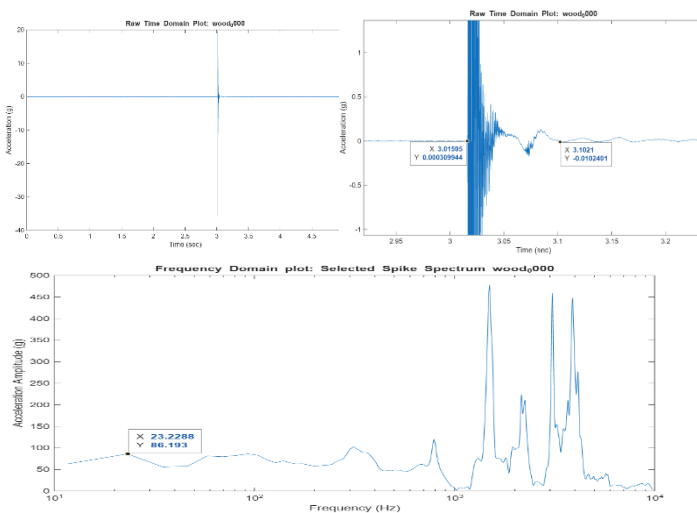


Fig. 12. Rap Test Time and frequency domain plots of Wood_0

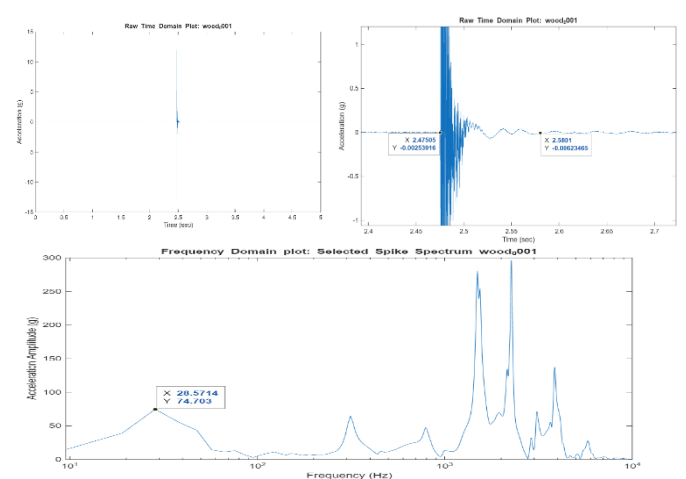


Fig. 13. Rap Test Time and frequency domain plots of Wood_1

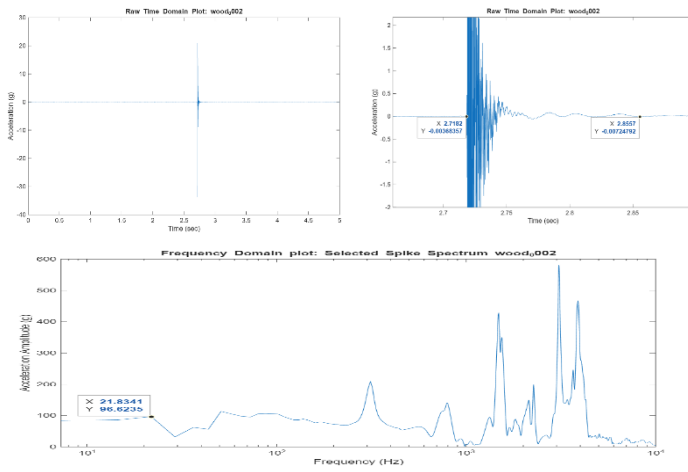


Fig. 14. Rap Test Time and frequency domain plots of Wood_2

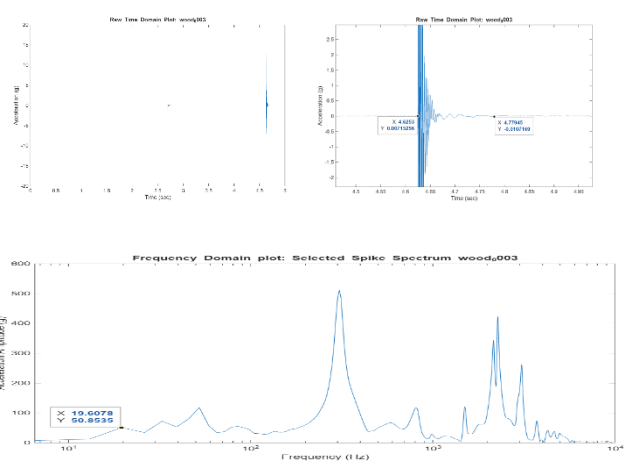


Fig. 15. Rap Test Time and frequency domain plots of Wood_3

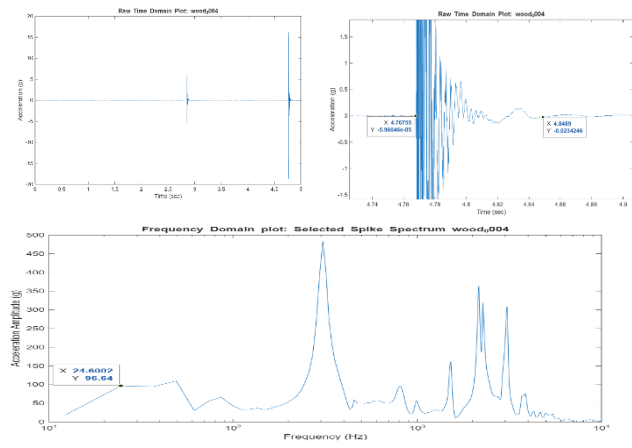


Fig. 16. Rap Test Time and frequency domain plots of Wood_4

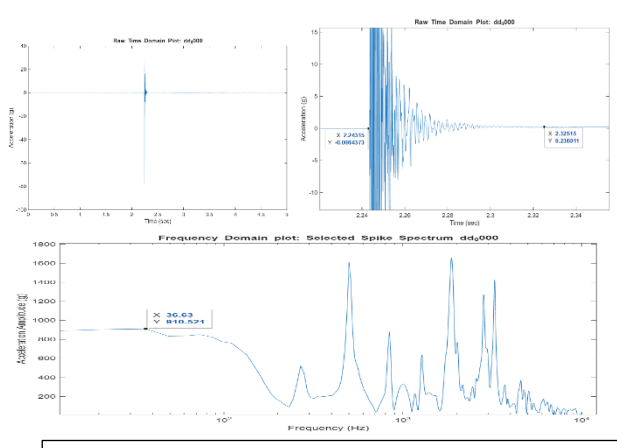


Fig. 17. Rap Test Time and frequency domain plots of DD_0

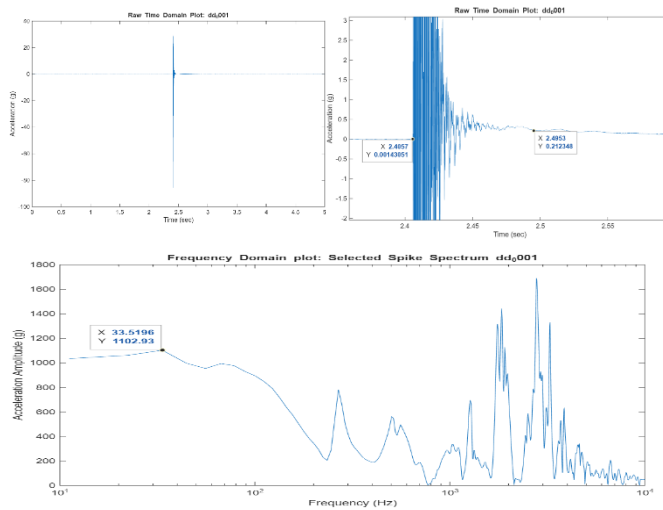


Fig. 18. Rap Test Time and frequency domain plots of DD_1

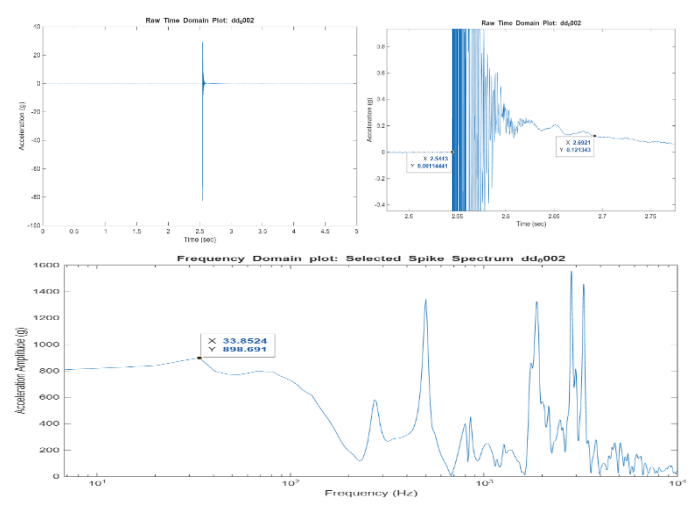


Fig. 19. Rap Test Time and frequency domain plots of DD_2

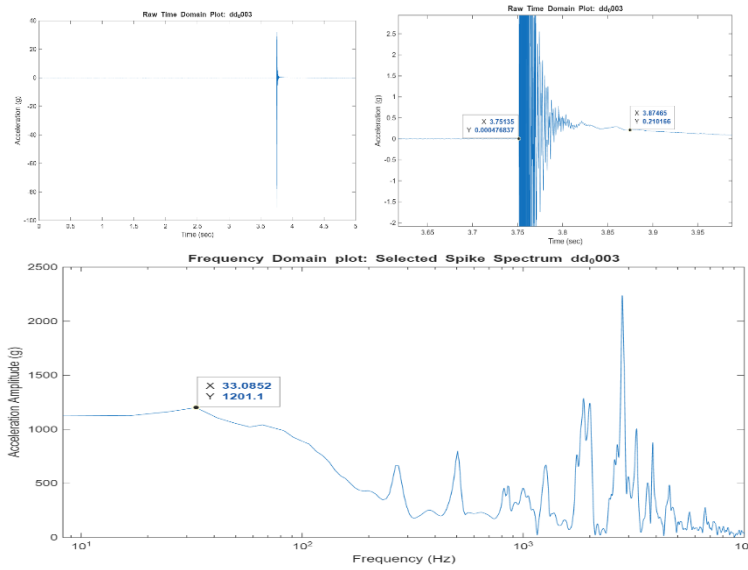


Fig. 20. Rap Test Time and frequency domain plots of DD_3

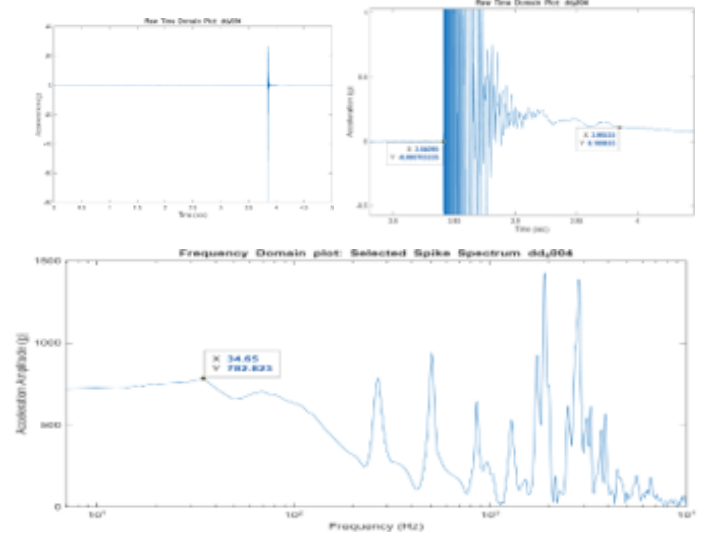


Fig. 21. Rap Test Time and frequency domain plots of DD_4

Material	RAP Frequency (Hz)	Material	RAP Frequency (Hz)	Material	RAP Frequency (Hz)	Material	RAP Frequency (Hz)
acrylic_0	41.2882	thermocol_0	34.722	wood_0	23.2288	dd_0	36.63
acrylic_1	43.6681	thermocol_1	36.9458	wood_1	28.5714	dd_1	33.5196
acrylic_2	40.404	thermocol_2	38.7679	wood_2	21.8341	dd_2	33.8524
acrylic_3	41.0678	thermocol_3	39.0795	wood_3	19.6078	dd_3	33.0852
acrylic_4	40.8719	thermocol_4	37.7358	wood_4	24.6002	dd_4	34.65

Table 3.

2.5 Drilling Experiments

Following RAP characterization, drilling trials were conducted on all four materials under identical conditions using a 5 mm drill bit, with five repetitions per material, yielding 20 datasets in total. Vibration signals were recorded for approximately 15 s per trial using the IEPE accelerometer and Dewesoft data acquisition system, and stored as CSV files.

Each recorded signal comprises four stages — idle, tool entry, steady-state drilling, and tool exit. Only the steady-state region, manually isolated by excluding entry and exit transients, was retained for analysis, since this segment alone reflects the true material-removal dynamics. This segment was additionally transformed via FFT to verify its dominant spectral components; however, the Health Indicator itself was developed from time-domain descriptors.

From the cropped time-domain signal, let the selected vibration signal be represented as x_i , where x_i denotes the vibration amplitude at the i^{th} sample and N is the total number of samples within the selected drilling interval. Seven physics-interpretable statistical features were extracted to characterize the drilling response, defined as follows.

Root Mean Square (RMS): The Root Mean Square represents the overall vibration energy contained in the signal.

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

Variance: Variance quantifies the dispersion of the vibration amplitudes around their mean value.

$$\sigma^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

where

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

Peak Value: Peak Value represents the maximum instantaneous vibration amplitude.

$$PV = \max(|x_i|)$$

Peak-to-Peak Value: Peak-to-Peak Value measures the total vibration excursion between the maximum and minimum amplitudes.

$$P2P = x_{max} - x_{min}$$

Crest Factor: Crest Factor describes the relationship between the highest impact amplitude and the overall vibration energy.

$$CF = \frac{PV}{RMS}$$

Skewness: Skewness measures the asymmetry of the vibration amplitude distribution.

$$Skewness = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^3}{\left(\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2} \right)^3}$$

Kurtosis: Kurtosis measures the impulsiveness or peakedness of the vibration signal and is particularly sensitive to transient events.

$$Kurtosis = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^4}{\left(\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 \right)^2}$$

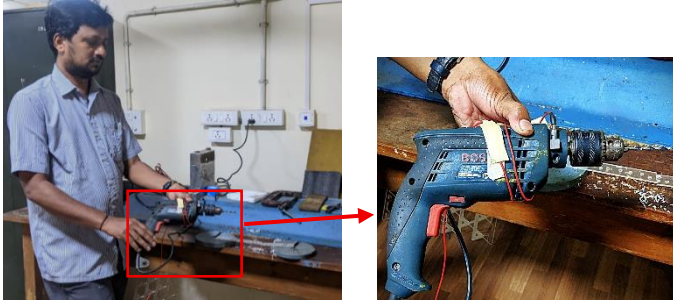


Fig. 22. Drilling machine, Accelerometer

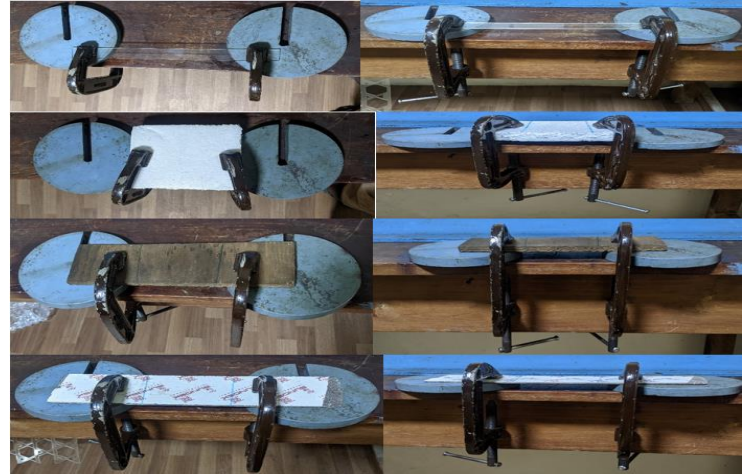


Fig. 23. Drilling Setup for, 4 materials

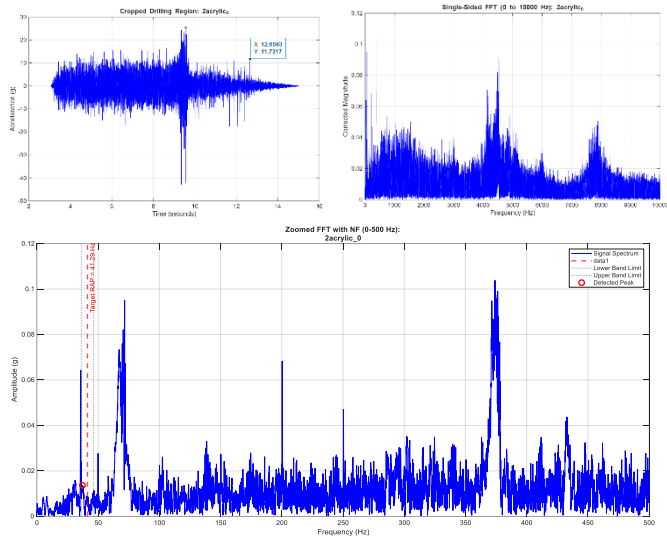


Fig. 24. Drilling Time and frequency domain plots of Acrylic_0

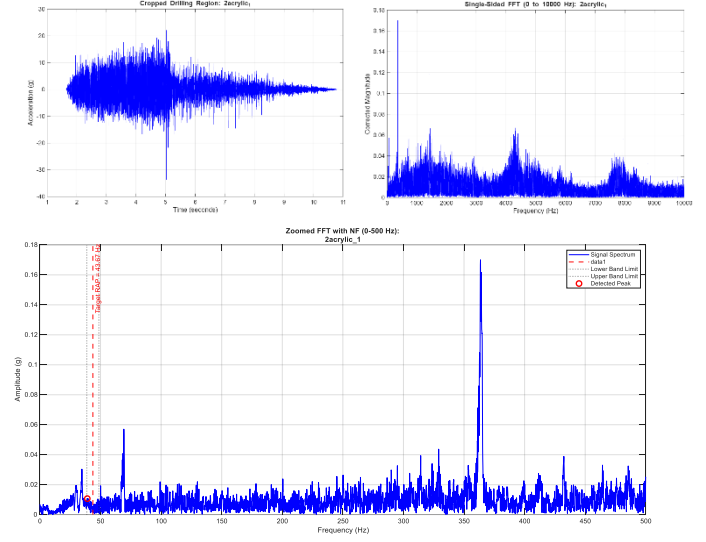


Fig. 25. Drilling Time and frequency domain plots of Acrylic_1

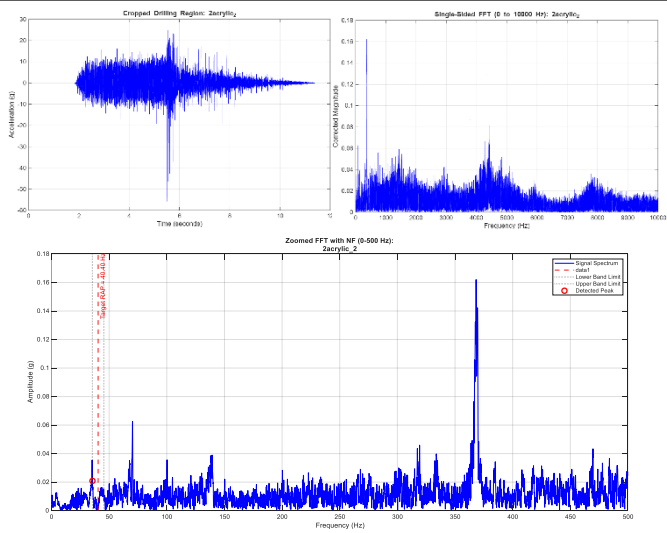


Fig. 26. Drilling Time and frequency domain plots of Acrylic_2

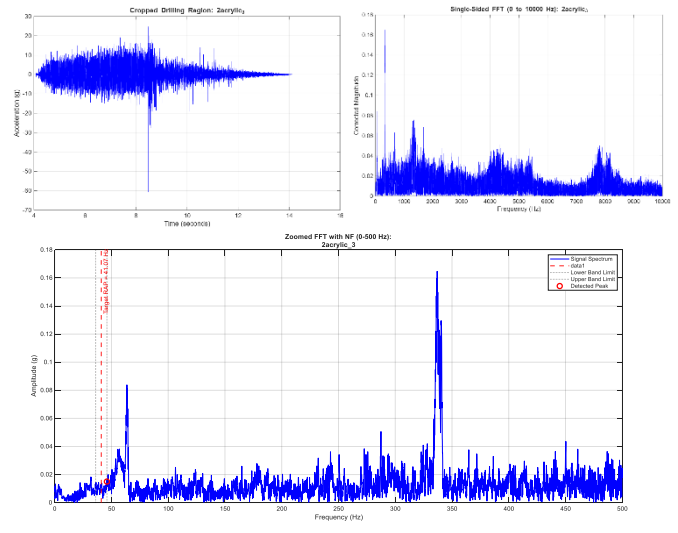


Fig. 27. Drilling Time and frequency domain plots of Acrylic_3

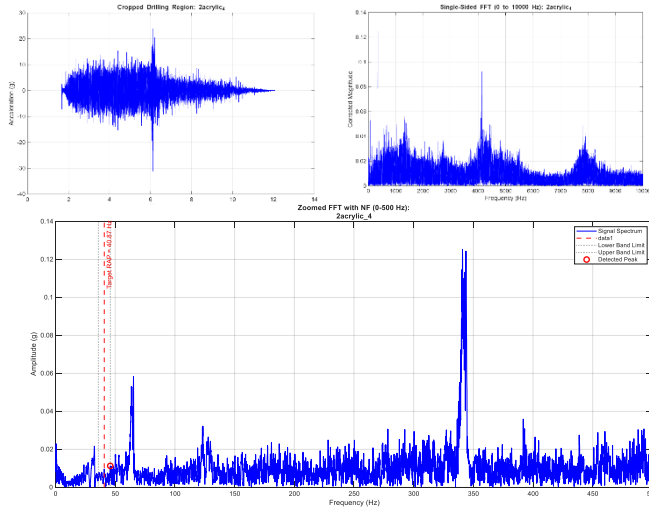


Fig. 28. Drilling Time and frequency domain plots of Acrylic_4

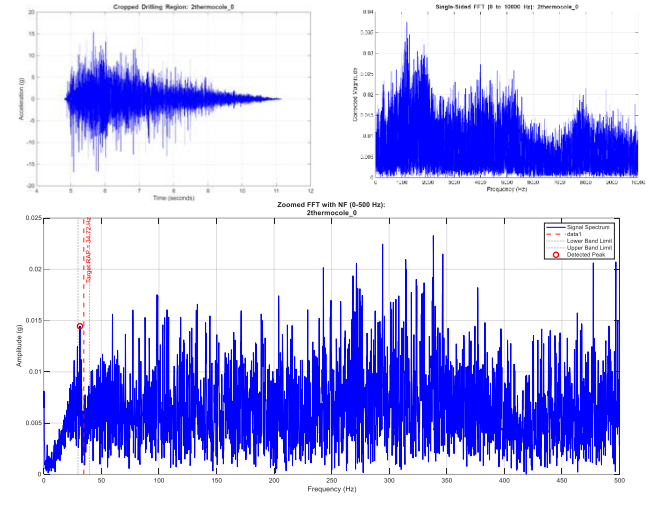


Fig. 29. Drilling Test Time and frequency domain plots of Thermocole_0

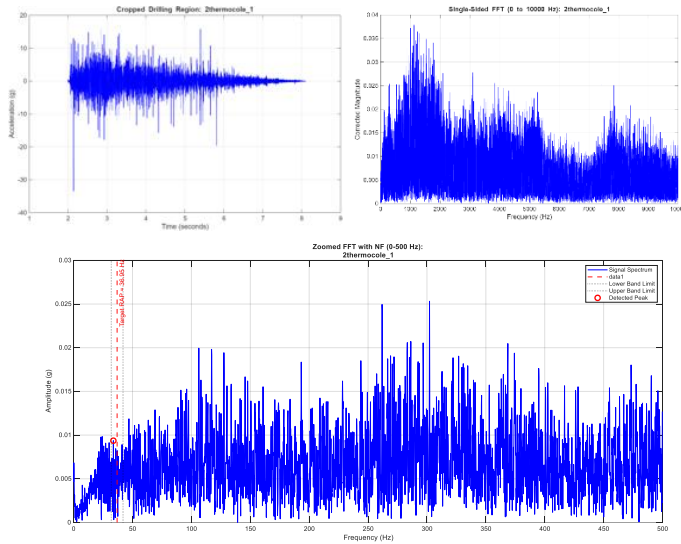


Fig. 30. Drilling Test Time and frequency domain plots of Thermocole_1

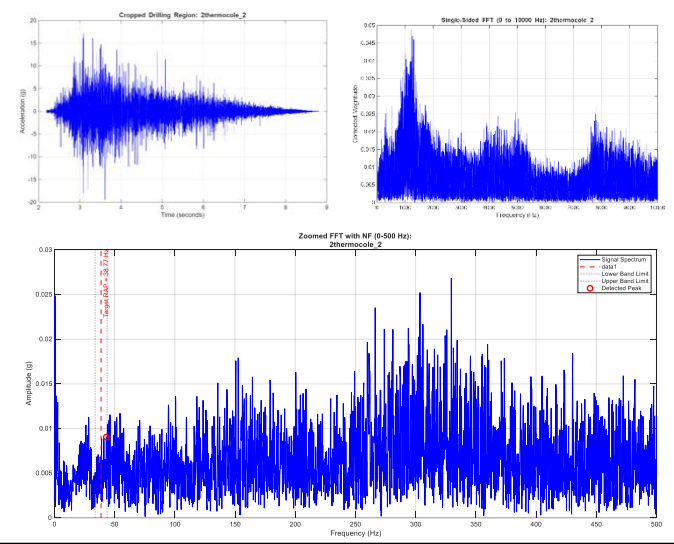


Fig. 31. Drilling Test Time and frequency domain plots of Thermocole_2

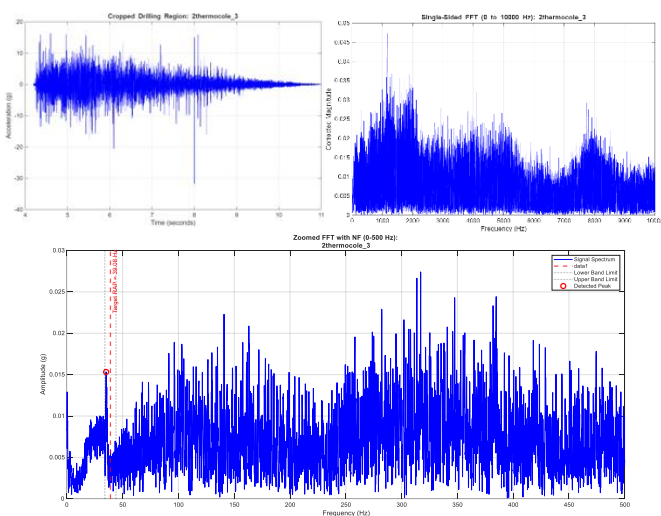


Fig. 32. Drilling Test Time and frequency domain plots of Thermocole_3

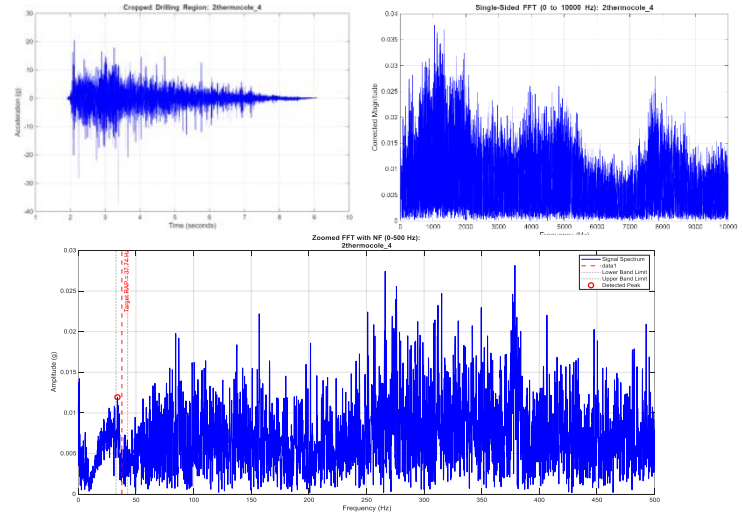


Fig. 33. Drilling Test Time and frequency domain plots of Thermocole_4

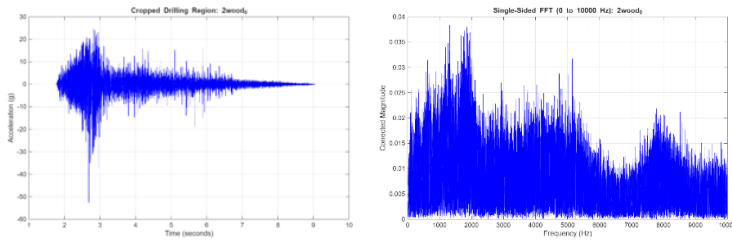


Fig. 34 Drilling Time and frequency domain plots of

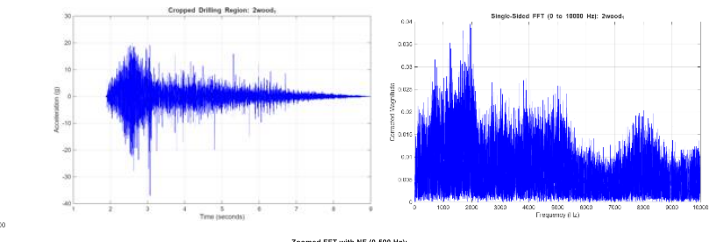


Fig. 35 Drilling Time and frequency domain plots of

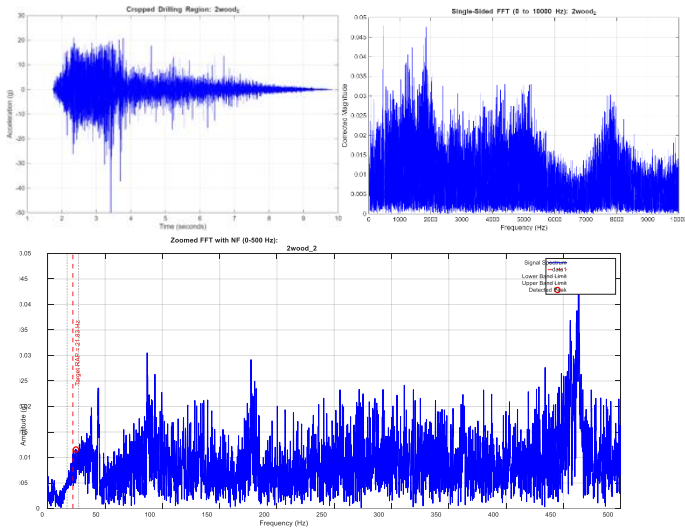


Fig. 36 Drilling Time and frequency domain plots of

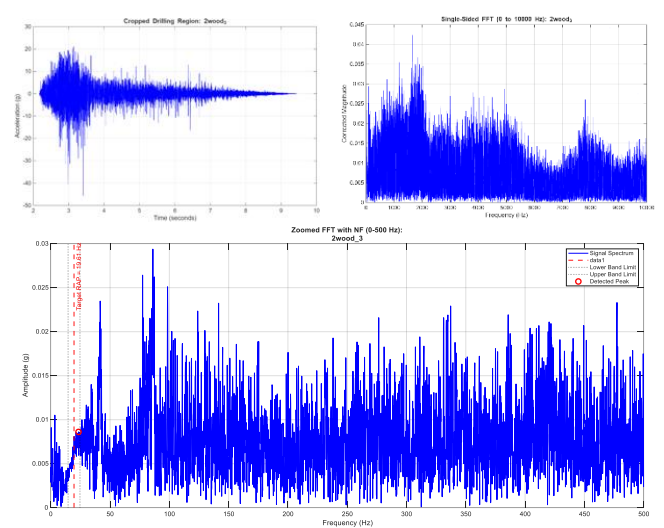


Fig. 37 Drilling Time and frequency domain plots of

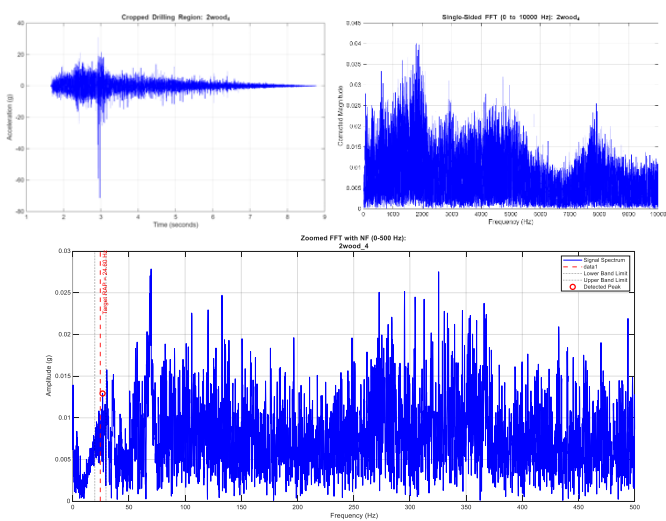


Fig. 38 Drilling Time and frequency domain plots of

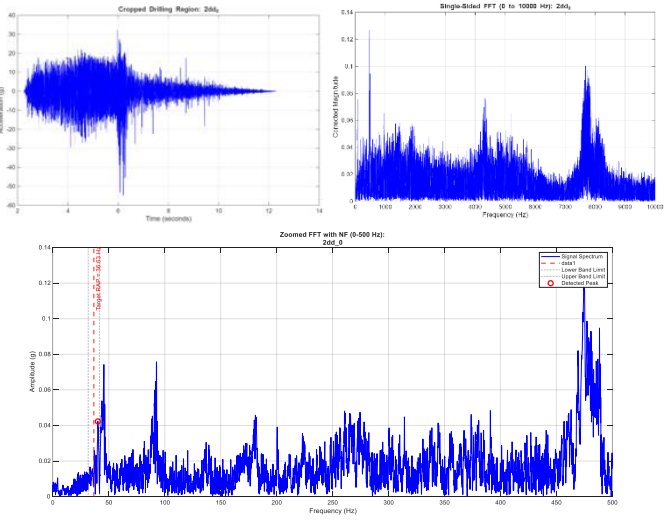


Fig. 39 Drilling Time and frequency domain plots of DD_0

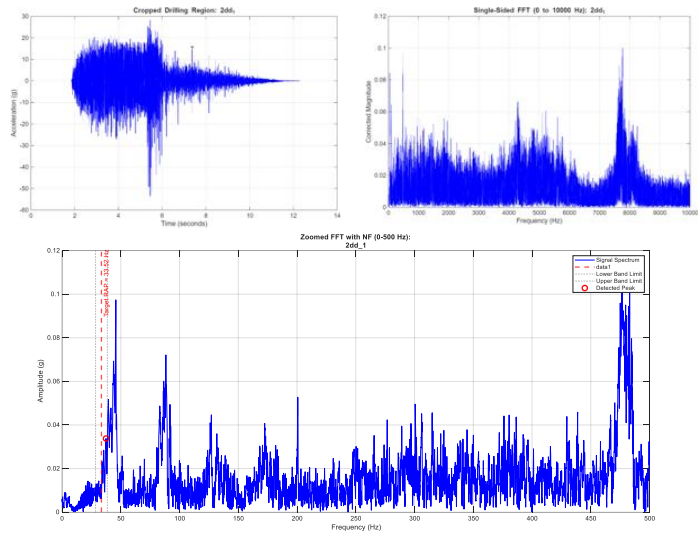


Fig. 40 Drilling Time and frequency domain plots of DD_1

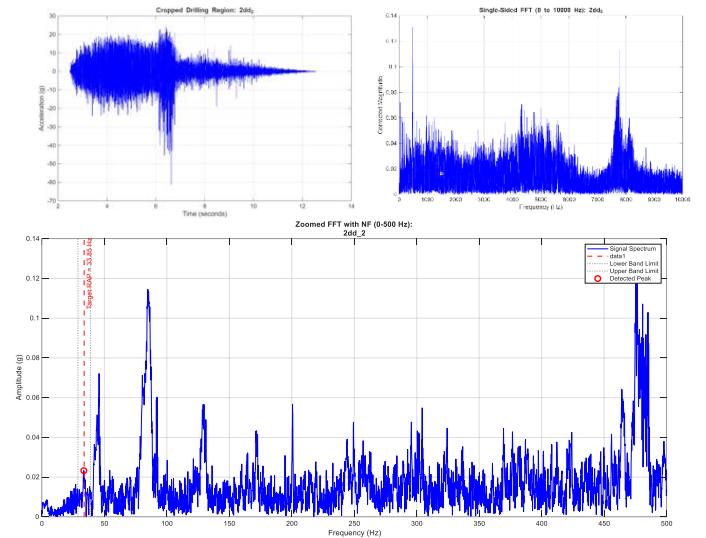


Fig. 41 Drilling Time and frequency domain plots of DD_2

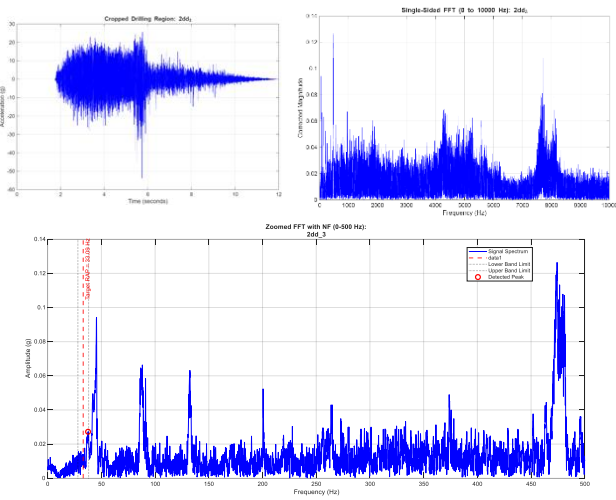


Fig. 42 Drilling Time and frequency domain plots of DD_3

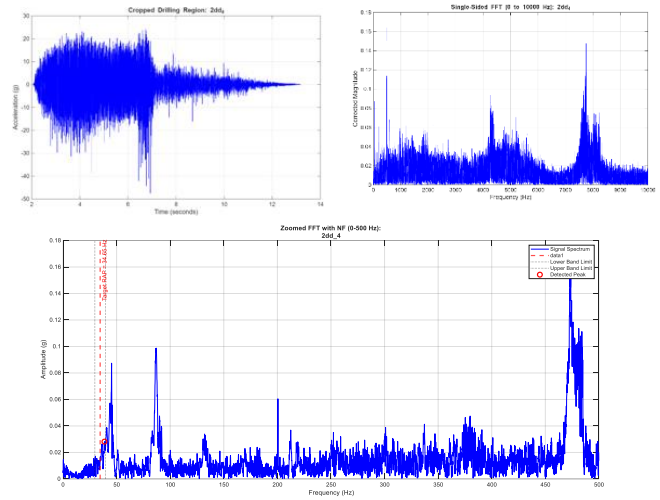


Fig. 43 Drilling Time and frequency domain plots of DD_4

3. Proposed Methodology

3.1 Physics-Guided Feature Screening Using Percentage Variation Analysis

Seven time-domain features were extracted from each drilling trial, namely Root Mean Square (RMS), Variance, Peak Value, Peak-to-Peak Value, Crest Factor, Kurtosis, and Skewness. Since these features represent different physical characteristics of the drilling vibration signal and have different numerical ranges and units, a direct comparison between them is not meaningful. Therefore, a percentage variation analysis was performed to evaluate all features on a common reference scale and identify the most suitable candidates for Health Indicator construction.

For each material, the feature value obtained from **Trial 0** was considered as the healthy baseline and assigned a normalized value of **100%**. The remaining trials were normalized with respect to this reference using

$$F_i(\%) = \frac{F_i}{F_0} \times 100$$

where F_i is the feature value of the i^{th} drilling trial and F_0 is the corresponding feature value obtained from Trial 0.

A normalized value of **100%** indicates no change from the healthy baseline, values greater than **100%** indicate an increase in feature magnitude, and values below **100%** indicate a decrease relative to the initial drilling condition.

To quantify the overall fluctuation of each feature across the repeated drilling trials, the **Variation Range Span** was calculated as

$$\Delta = \max(F\%) - \min(F\%)$$

where Δ represents the total percentage variation exhibited by a feature during the drilling experiments.

A small value of Δ indicates that the feature remains stable and repeatable across repeated drilling trials of the same material. A large value of Δ indicates that the feature exhibits greater sensitivity to drilling progression or material changes. However, excessively large fluctuations may also indicate poor repeatability, making the feature less suitable for Health Indicator construction.

Each feature was evaluated using four engineering criteria:

- - Stability during repeated drilling experiments.
- - Sensitivity to drilling progression from the first to the fifth drilling trial.
- - Ability to distinguish the vibration behaviour of different materials.
- - Physical interpretability with respect to the drilling process.

The normalized feature trajectories for all four materials are presented in **Figure 3.2**.

Among the extracted features, **RMS** and **Variance** exhibited consistent and stable behaviour across the five drilling trials for all materials, indicating good repeatability while effectively representing the overall vibration energy and its dispersion. **Peak Value** showed a strong response to drilling progression and material interaction by capturing the maximum impact events occurring during drilling. **Kurtosis** remained highly responsive for soft and heterogeneous materials, demonstrating its ability to detect impulsive responses associated with localized fracture and structural changes.

In contrast, **Peak-to-Peak Value** provided information similar to that already represented by Peak Value and therefore did not contribute additional physical insight. **Crest Factor** is mathematically derived from Peak Value and RMS. Since both constituent features were retained independently, Crest Factor was considered redundant. **Skewness** remained close to zero for all materials throughout the repeated drilling trials, indicating nearly symmetric vibration amplitude distributions and limited capability for distinguishing changes in drilling behaviour.

Based on the percentage variation analysis and the engineering interpretation of the extracted features, **RMS, Variance, Peak Value, and Kurtosis** were selected for subsequent Health Indicator development. These four features collectively represent the overall vibration energy, energy dispersion, maximum impact response, and signal impulsiveness, providing complementary information about the drilling process.

Unlike conventional feature selection approaches that rely primarily on statistical ranking or machine learning algorithms, the proposed feature screening process combines quantitative percentage variation analysis with engineering interpretation to identify features that are both physically meaningful and robust for Health Indicator construction.

The selected features are subsequently assigned objective weights using the **Entropy Weight Method (EWM)**, which is described in the following section.

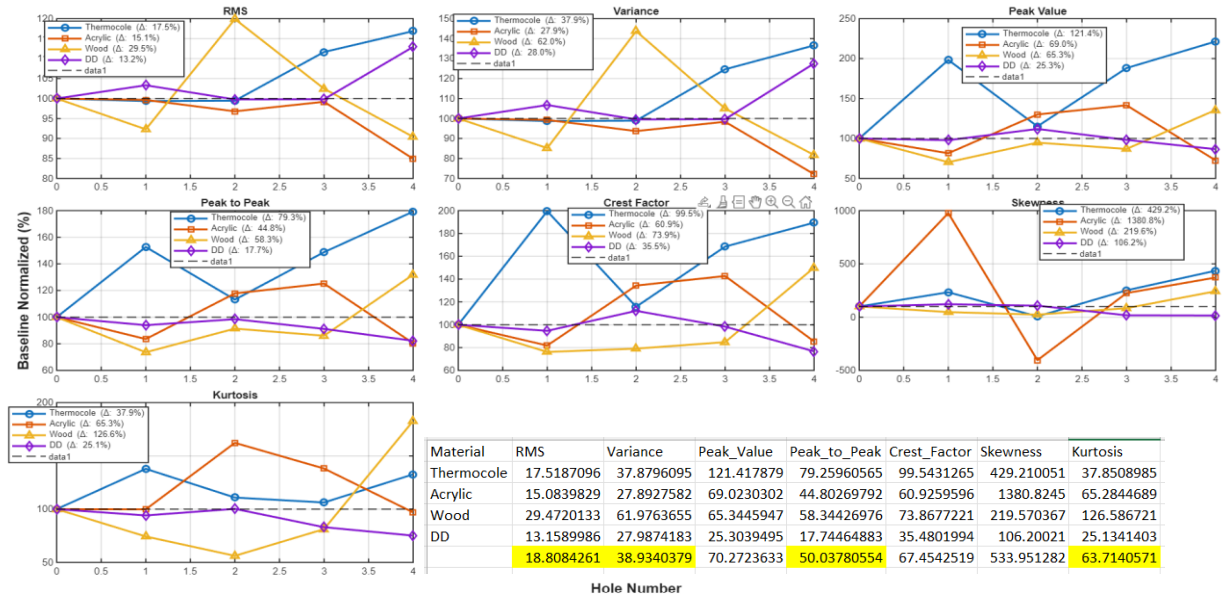


Fig. 44. Percentage Variation plots for Time Domine feature selection

3.2 Entropy Weight Method (EWM) for Objective Feature Weighting

The four selected features, namely RMS, Variance, Peak Value, and Kurtosis, describe different physical aspects of the drilling process and do not contribute equally to representing the drilling condition. Therefore, an objective weighting strategy is required before combining them into a single representative index.

The Entropy Weight Method (EWM) is adopted to assign feature weights based on Shannon's Information Entropy. Features that exhibit greater variation across the observation windows contain more diagnostic information and receive higher weights. Conversely, features that remain relatively uniform contribute less information and therefore receive lower weights. This approach eliminates subjective judgment during the feature weighting process.

Step 1: Min-Max Normalization

Each feature is first normalized to the range of 0 to 1 using

$$z_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)}$$

where x_{ij} is the value of feature j at observation i , and z_{ij} is the normalized feature value. This normalization ensures that no feature dominates the weighting process because of its numerical magnitude.

Step 2: Probability Distribution

The normalized feature values are converted into a probability distribution using

$$p_{ij} = \frac{z_{ij}}{\sum_{i=1}^n z_{ij}}$$

where p_{ij} represents the probability associated with the normalized value of feature j .

Step 3: Information Entropy

The entropy of each feature is calculated as

$$E_j = -k \sum_{i=1}^n p_{ij} \ln(p_{ij}), k = \frac{1}{\ln(n)}$$

where n is the total number of observations. A feature with high entropy exhibits nearly uniform behaviour across the observation windows and therefore contains limited diagnostic information. In contrast, a feature with low entropy exhibits greater variation and provides more useful information for distinguishing different drilling conditions.

Step 4: Feature Weight Calculation

The information utility of each feature is calculated as

$$D_j = 1 - E_j$$

The objective weight assigned to each feature is then obtained using

$$w_j = \frac{D_j}{\sum_{j=1}^m D_j}$$

where m is the total number of selected features. The calculated weights satisfy

$$\sum_{j=1}^m w_j = 1$$

ensuring that each feature contributes in proportion to its information content.

The computed feature weights are subsequently used to combine the four selected features into a single **Feature Fusion Index (FFI)**, which forms the basis for the proposed Health Indicator construction described in the following section.

3.3 Feature Fusion Index (FFI) Construction

The four selected features, namely RMS, Variance, Peak Value, and Kurtosis, represent different physical characteristics of the drilling vibration signal. Since these features contribute differently to the drilling condition, they are combined using the objective weights obtained from the Entropy Weight Method to construct a single **Feature Fusion Index (FFI)**.

The FFI for the k^{th} observation window is calculated as

$$FFI(k) = \sum_{j=1}^m w_j f_j(k)$$

where

- $f_j(k)$ is the normalized value of the j^{th} feature at window k ,
- w_j is the entropy weight assigned to the corresponding feature,
- m is the number of selected features.

The resulting FFI combines the information from vibration energy (RMS), energy dispersion (Variance), maximum impact response (Peak Value), and signal impulsiveness (Kurtosis) into a single dimensionless index representing the instantaneous drilling condition.

The FFI is treated as the observation input to the Kalman Filter, which estimates the underlying Health Indicator by reducing measurement noise while preserving the true degradation trend.

3.4 Kalman Filter-Based Health Indicator Construction

Although the Feature Fusion Index (FFI) combines the selected features into a single representative index, it still contains measurement noise arising from sensor disturbances and random vibration fluctuations. Therefore, a Kalman Filter is employed to estimate the underlying drilling health condition from the noisy FFI measurements.

The Kalman Filter models the drilling process as a dynamic system whose hidden state consists of the drilling Health Indicator and its rate of change. The state vector is defined as

$$X(k) = \begin{bmatrix} h(k) \\ \dot{h}(k) \end{bmatrix}$$

where

- $h(k)$ is the estimated drilling Health Indicator.
- $\dot{h}(k)$ is the Health Evolution Rate (HER).

The state prediction is performed using

$$\hat{X}_{k|k-1} = A\hat{X}_{k-1|k-1}$$

where the state transition matrix is

$$A = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$$

The measured Feature Fusion Index is used as the observation input,

$$z(k) = FFI(k)$$

The predicted state is then updated using the Kalman correction equation,

$$\hat{X}_{k|k} = \hat{X}_{k|k-1} + K_k(z_k - H\hat{X}_{k|k-1})$$

where K_k is the Kalman gain and H is the observation matrix.

The Kalman Filter recursively combines the predicted health state with the measured FFI to suppress random measurement noise while preserving the underlying degradation trend. The estimated health state $h(k)$ is considered as the proposed **Health Indicator (HI)**, while its rate of change $\dot{h}(k)$ is referred to as the **Health Evolution Rate (HER)**. These two parameters are subsequently used for material transition and breakthrough detection.

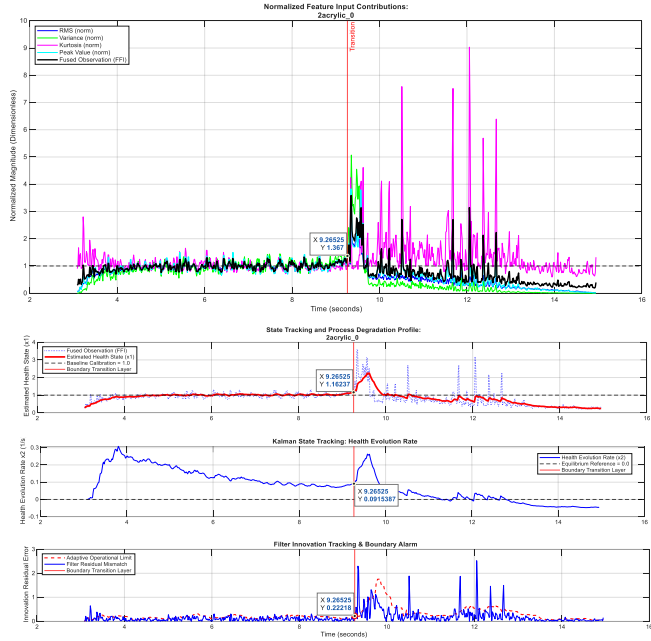


Fig. 45 Health Indicator plots of Acrylic_0

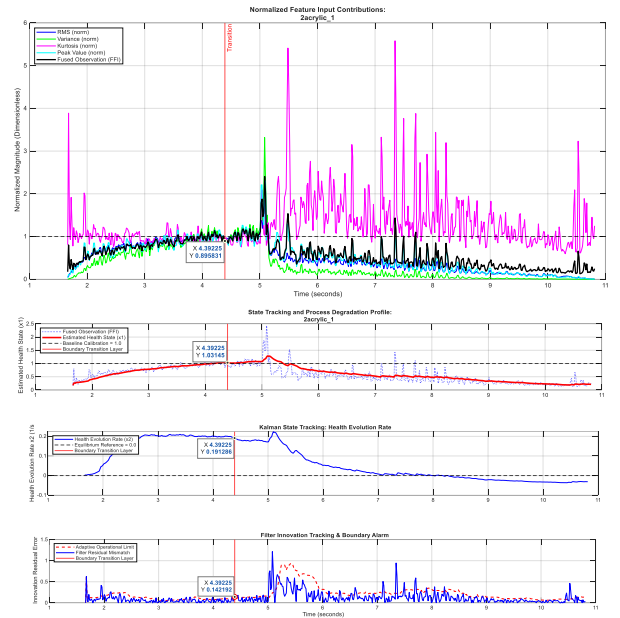


Fig. 46 Health Indicator plots of Acrylic_1

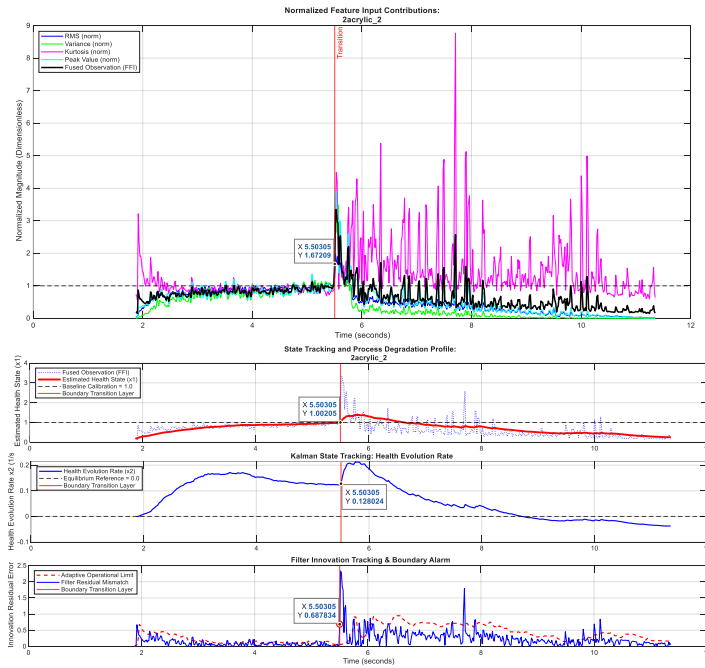


Fig. 47 Health Indicator plots of Acrylic_2

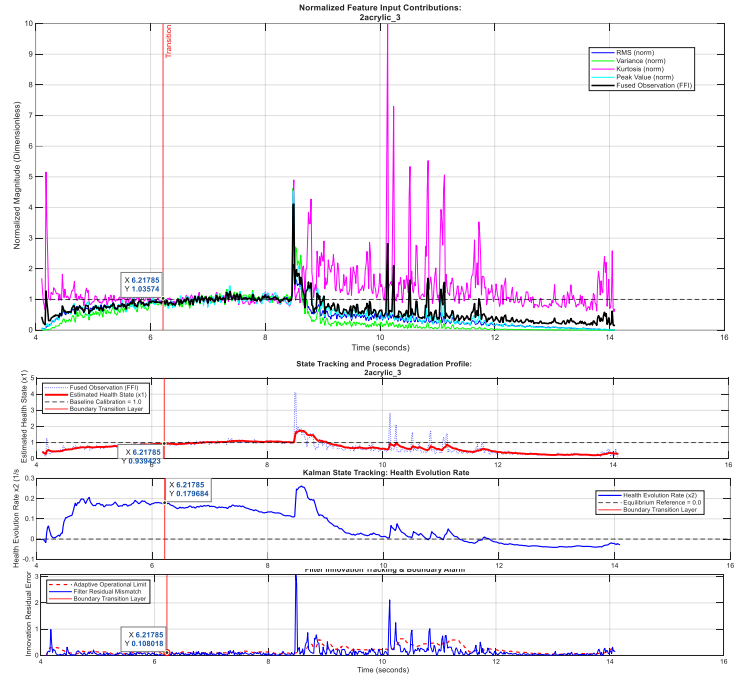


Fig. 48 Health Indicator plots of Acrylic_3

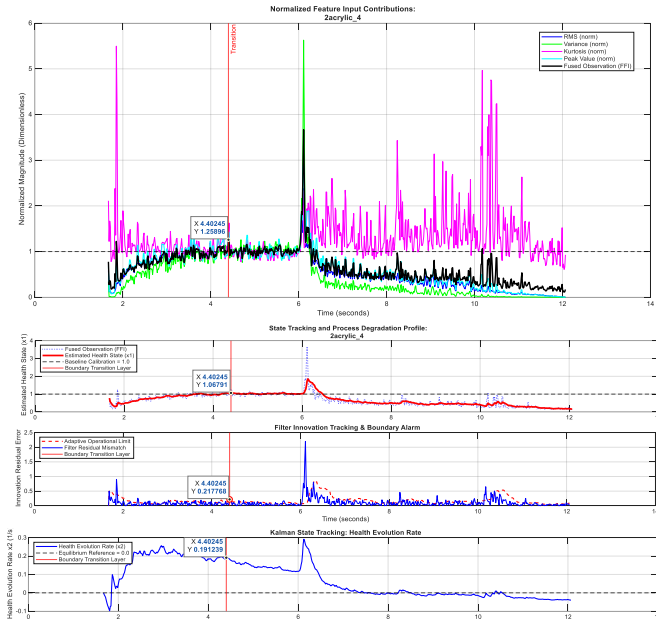


Fig. 49 Health Indicator plots of Acrylic_4

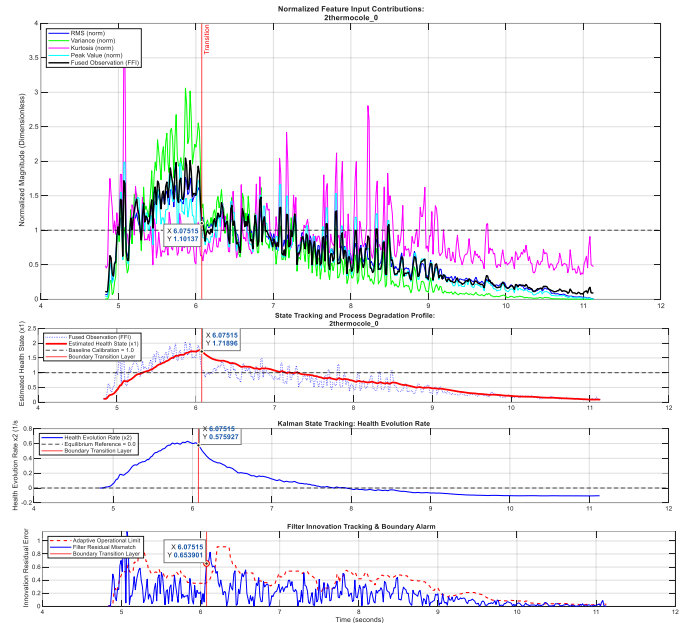


Fig. 50 Health Indicator plots of Theromocole_0

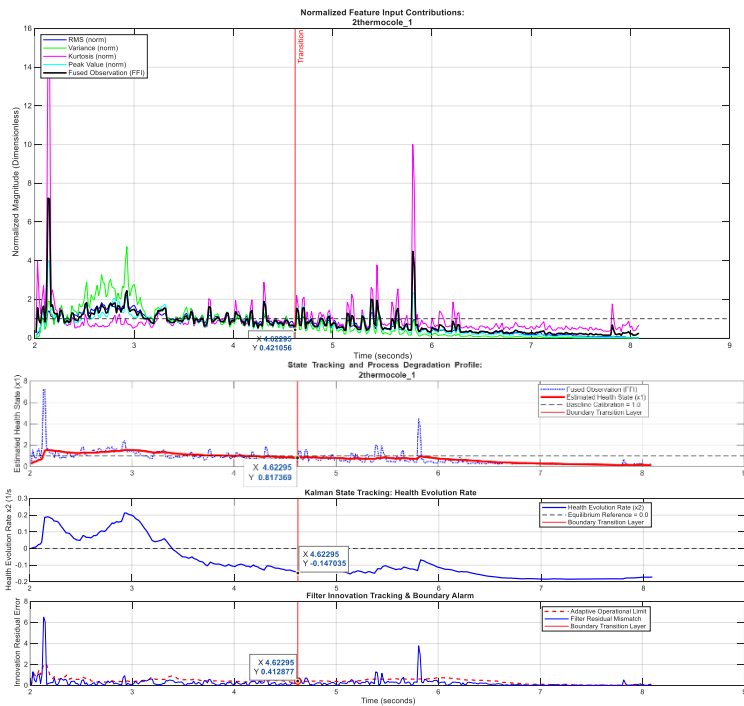


Fig. 51 Health Indicator plots of Theromocole_1

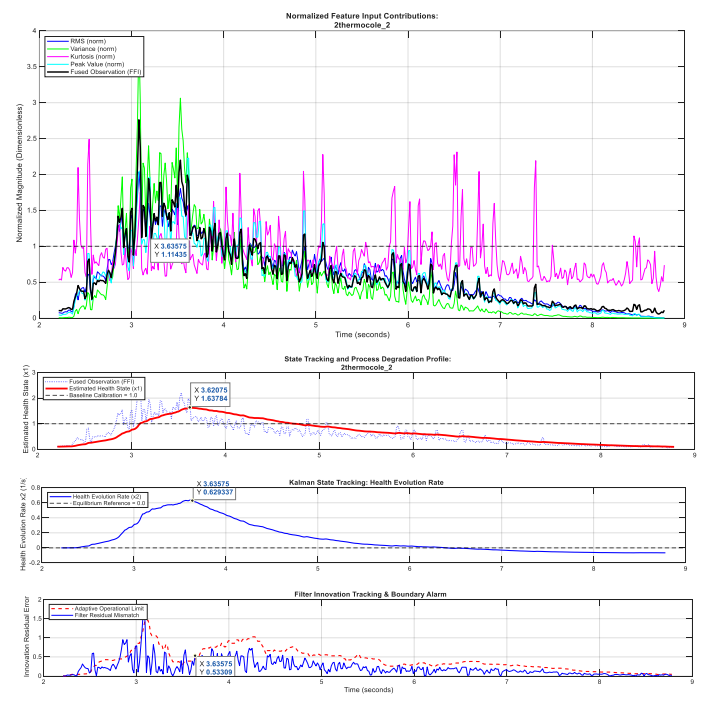


Fig. 52 Health Indicator plots of Theromocole_2

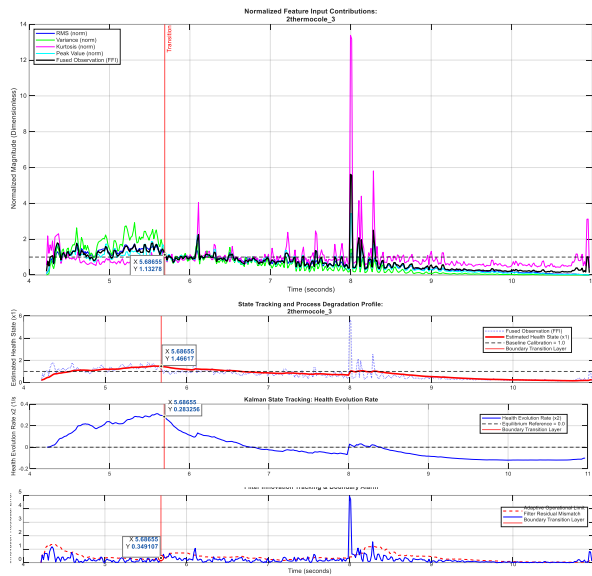


Fig. 53 Health Indicator plots of Theromocole_3

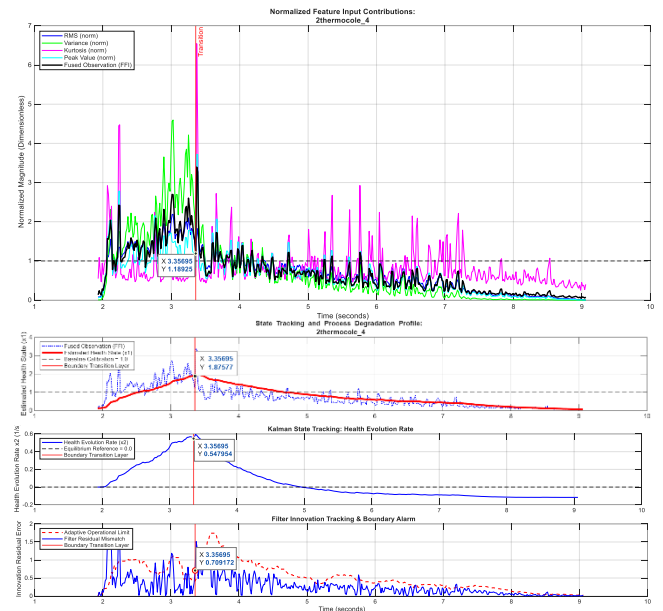


Fig. 54 Health Indicator plots of Theromocole_4

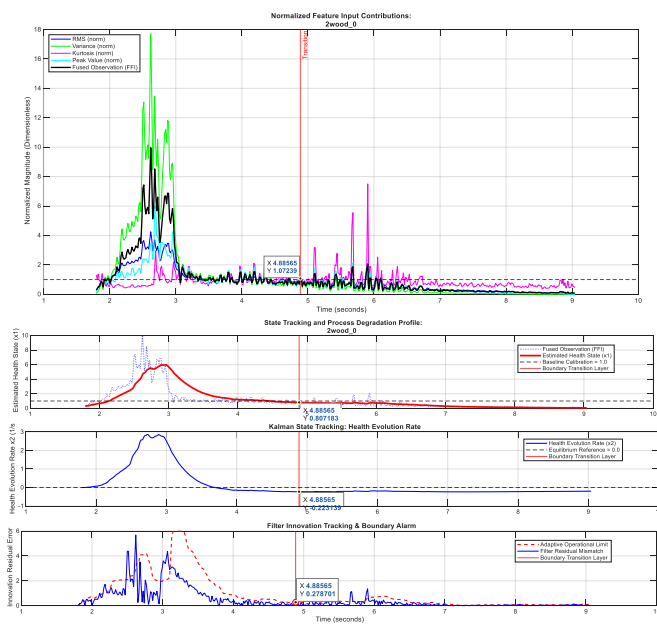


Fig. 55 Health Indicator plots of Wood_0

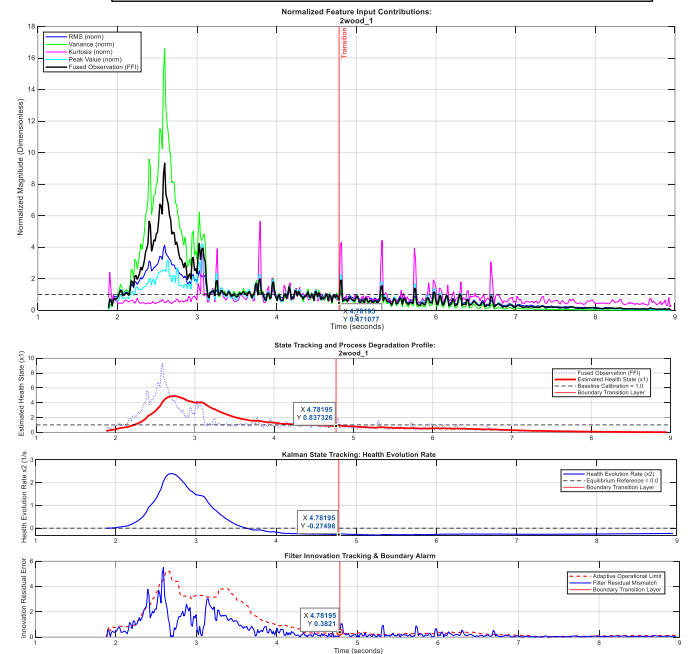


Fig. 56 Health Indicator plots of Wood_1

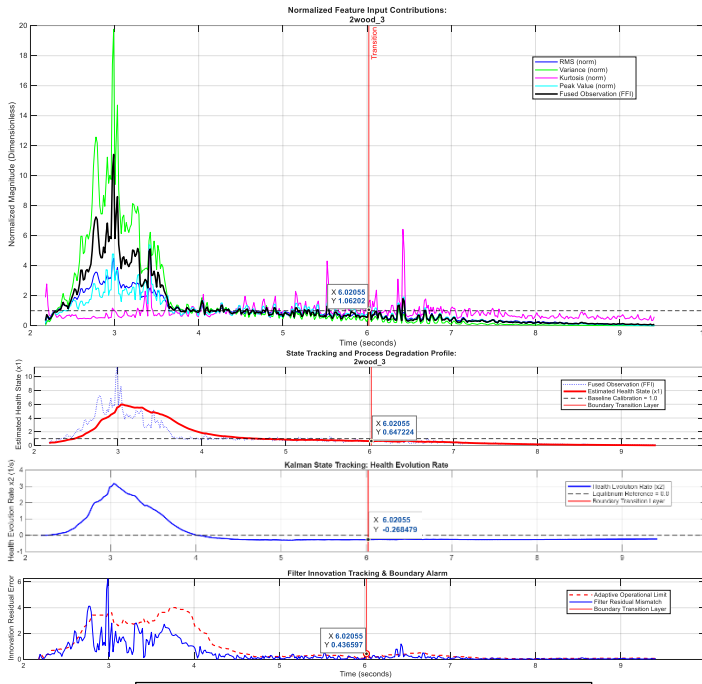


Fig. 57 Health Indicator plots of Wood_2

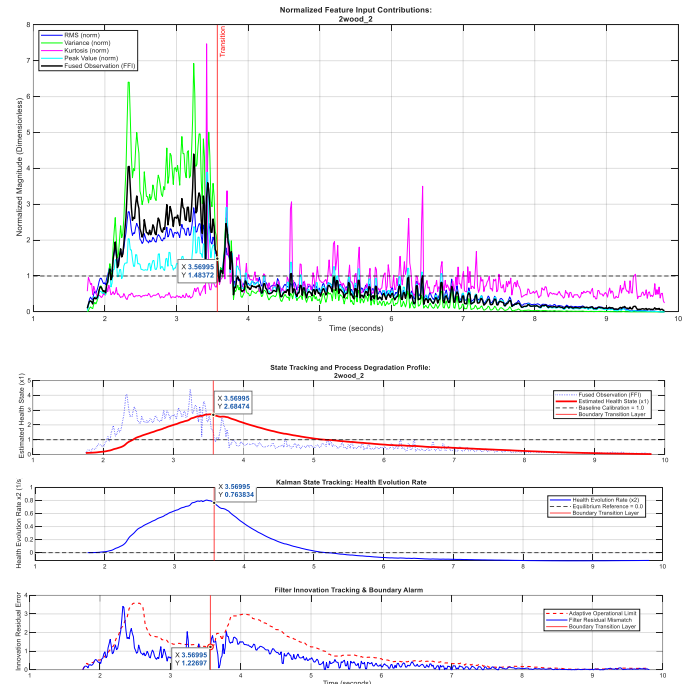


Fig. 58 Health Indicator plots of Wood_3

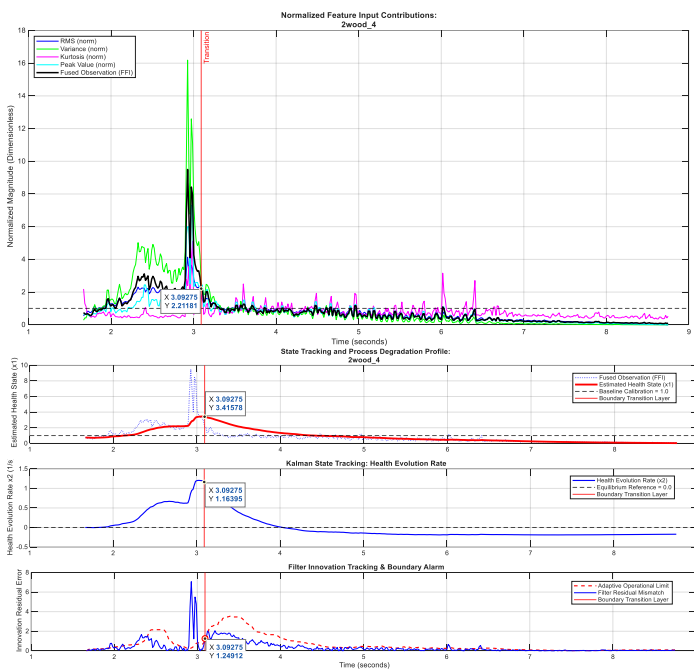


Fig. 59 Health Indicator plots of Wood_4

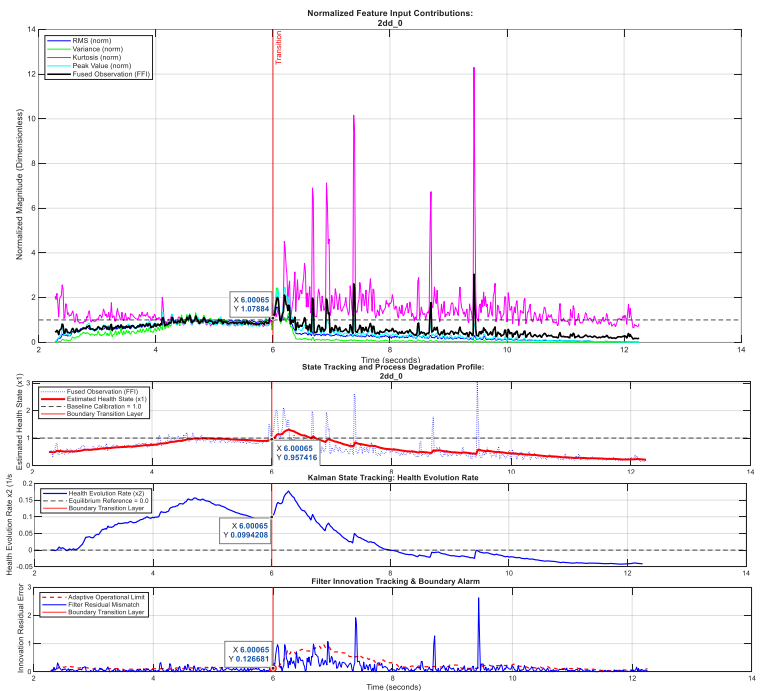


Fig. 60 Health Indicator plots of DD_0

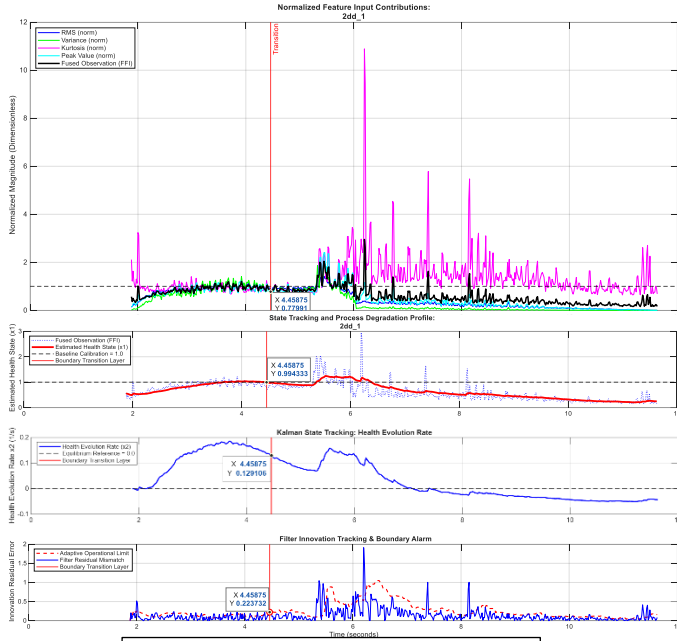


Fig. 61 Health Indicator plots of DD_1

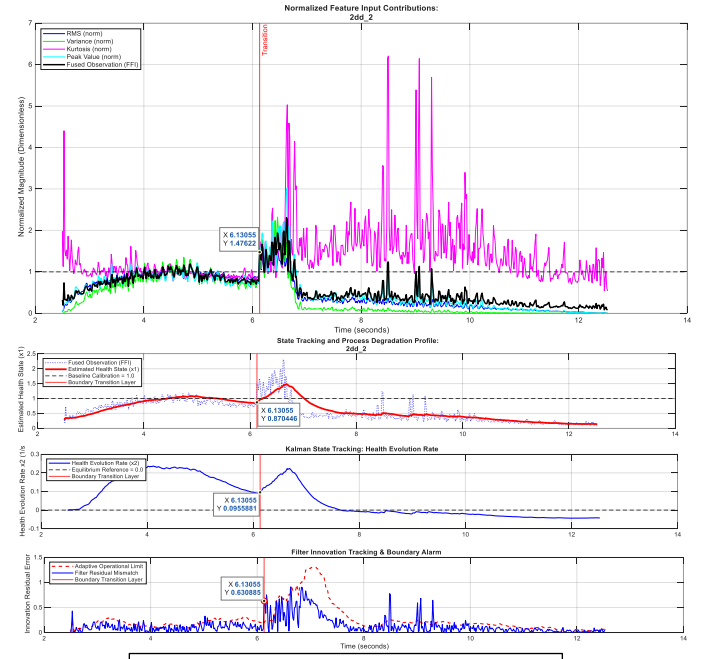


Fig. 62 Health Indicator plots of DD_2

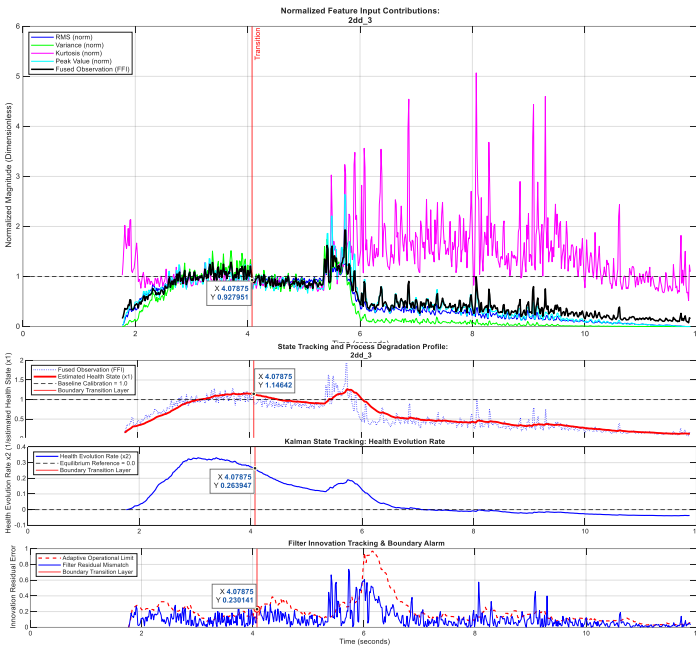


Fig. 63 Health Indicator plots of DD_3

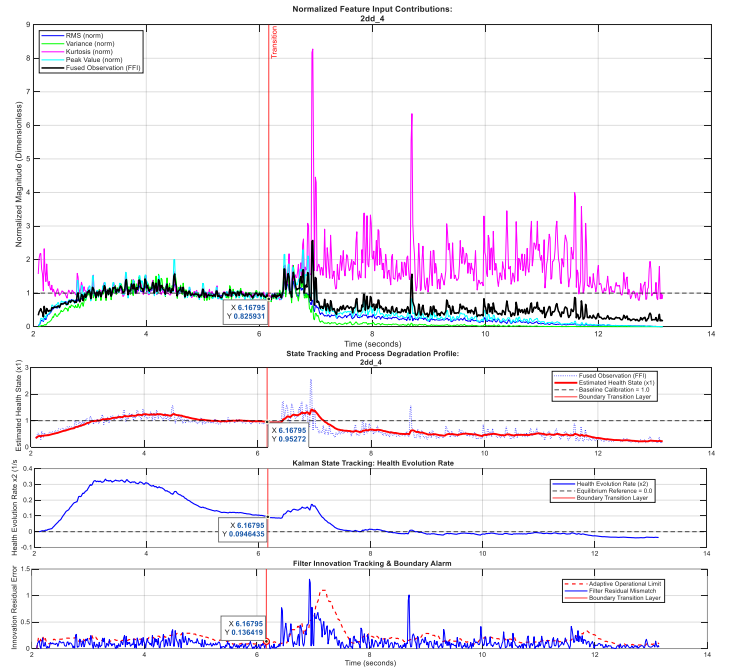


Fig. 64 Health Indicator plots of DD_4

4. Results and Discussion

4.1 Health Indicator Performance

The proposed Health Indicator was evaluated using all twenty drilling experiments. Across all materials, the Health Indicator exhibited a consistent evolution throughout the drilling process. The Kalman filter effectively suppressed the fluctuations present in the Feature Fusion Index while preserving the underlying drilling behavior. The Health Evolution Rate and innovation residual responded to changes in drilling conditions, enabling the transition between different drilling stages to be clearly identified. The consistent response observed across repeated trials indicates that the proposed framework captures the physical drilling process rather than random measurement noise.

4.2 Validation of the Health Indicator

The Health Indicator was validated across twenty drilling experiments using four criteria: repeatability, trendability, monotonicity, and physical consistency.

Repeatability was evaluated using the coefficient of variation (CV) of the mean Health Indicator across five trials of each material:

$$CV = \frac{\sigma}{\mu} \times 100\%$$

where σ is the standard deviation and μ is the mean of the five trial values. A CV below 15% indicates good repeatability. The validation results show that DD and thermocol achieved CV values of approximately 7.6%, while acrylic and wood achieved 12.8% and 14.6%, respectively. All four materials fall within the acceptable range, confirming that the proposed framework produces consistent Health Indicator estimates under repeated drilling of the same material.

Trendability was assessed by examining whether the Health Indicator metrics drift systematically from trial 0 to trial 4. The Pearson correlation coefficient between trial index and mean Health Indicator was computed for each material. Values close to zero indicate no systematic drift across repeated trials. The results showed no significant trend in any material, confirming that the Health Indicator does not degrade over successive drilling experiments.

Monotonicity was evaluated using the global Health Indicator change:

$$\Delta HI = HI_{\text{end}} - HI_{\text{start}}$$

A negative value indicates that the Health Indicator declines overall during drilling, which is the physically expected behavior because the drill progressively disengages from the material as it approaches and exits the specimen. The majority of trials across all materials showed a negative change, confirming the expected monotonic tendency. The exceptions were a small number of trials where bench contact after breakthrough temporarily elevated the Health Indicator before it returned to zero.

Physical consistency was investigated by comparing the mean Health Indicator and maximum Health Evolution Rate against independently measured material properties, density and RAP natural frequency. The results show that materials with higher density and higher natural frequency produced higher peak Health Indicator values and stronger Health Evolution Rate peaks during drilling. This is physically expected, since denser and stiffer materials resist the drill more strongly, generating higher vibration energy and more pronounced structural dynamic responses. The consistency between the Health Indicator behavior and these independently measured properties confirms that the estimated Health Indicator reflects genuine material physics rather than an artifact of the signal processing chain.

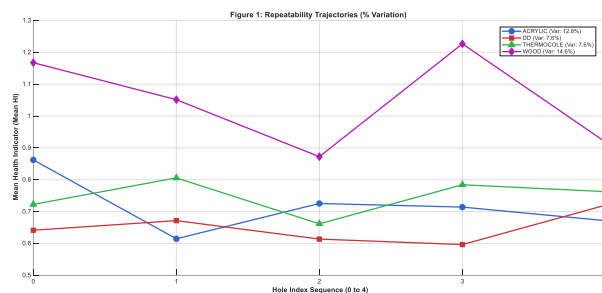


Fig. 65. Repeatability Analysis

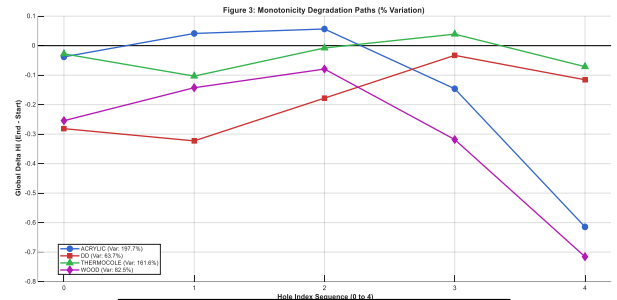


Fig. 66. Monotonicity Analysis

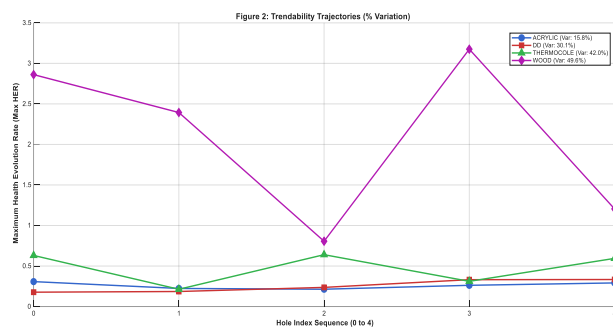


Fig. 67. Trendability Analysis

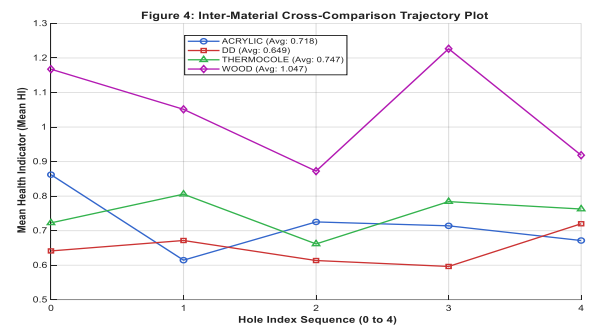


Fig. 68. Inter-Material, HI Comparison

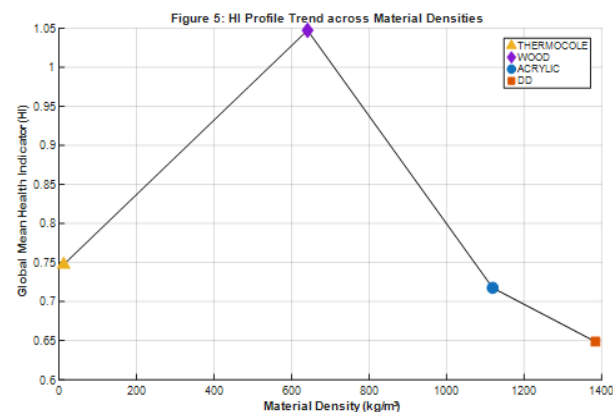


Fig. 69. HI vs Densities

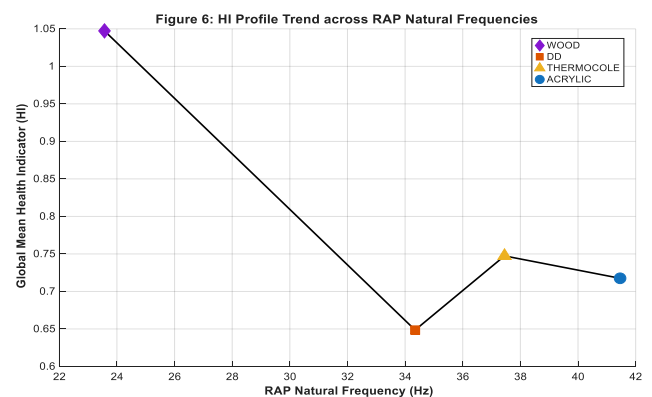


Fig. 70. HI vs Natural Frequency

4.3 Proof-of-Concept Validation Using Combined Material Signals

To investigate the capability of the proposed framework for detecting material transitions, a synthetic drilling signal was created by concatenating the steady-state drilling signals of two different materials.

Without modifying any model parameters, the proposed framework identified a clear transition between the two segments. The Health Indicator, Health Evolution Rate, and innovation residual all responded distinctly at the transition point, demonstrating sensitivity to changes in material characteristics.

Although the combined signal does not fully represent a real layered specimen, the results provide initial evidence that the proposed Health Indicator can detect changes in drilling conditions associated with different materials. Experimental validation on actual multilayer specimens will be carried out in future work.

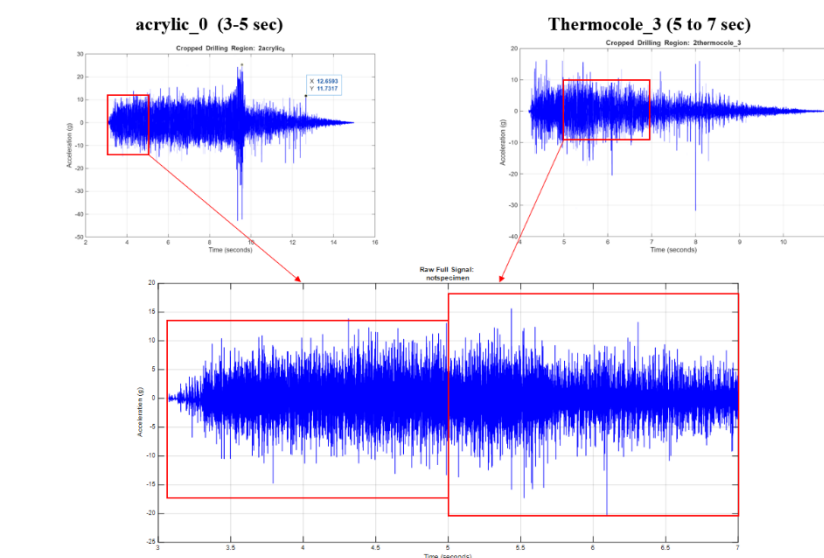


Fig. 71. Synthetic Data Construction

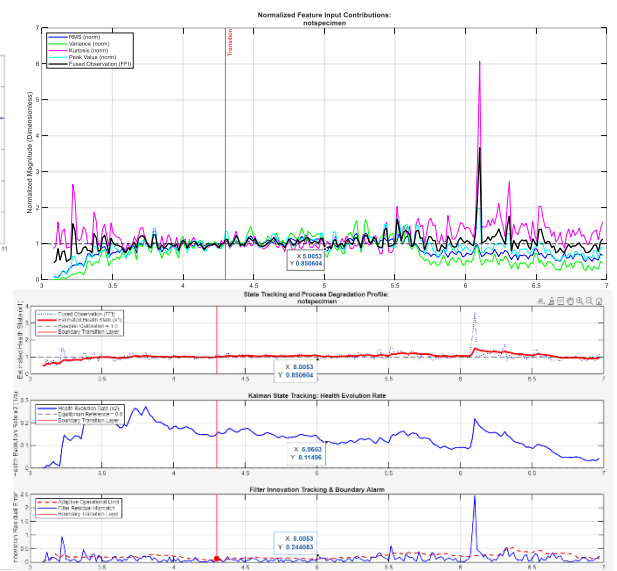


Fig. 72. Health indicator plots of Synthetic Data

4.4 Discussion

The results show that the proposed framework provides a stable and physically interpretable Health Indicator using only vibration measurements. Feature selection based on percentage variation analysis, objective weighting through the Entropy Weight Method, and state estimation using the Kalman filter collectively contributed to the robustness of the framework.

The agreement between the Health Indicator and independently measured material properties, together with the successful proof-of-concept validation on combined material signals, indicates that the proposed approach is capable of representing genuine changes in drilling conditions. These results provide a strong basis for extending the framework to real multilayer drilling experiments.

5. Conclusion

A physics-guided Health Indicator framework for material state identification and transition monitoring during drilling was developed using vibration analysis. The proposed framework integrates physics-guided feature selection, entropy-based feature weighting, Feature Fusion Index (FFI) construction, and Kalman filter state estimation to generate a mathematically defined and physically interpretable Health Indicator.

The framework was validated using twenty drilling experiments performed on four engineering materials. The obtained results demonstrated consistent Health Indicator behaviour with good repeatability, trendability, monotonicity, and agreement with independently measured material properties. The proof-of-concept study using synthetically combined material signals further demonstrated the capability of the proposed framework to detect changes in material state.

Although the proposed Health Indicator successfully captured drilling progression and material transitions, the selected time-domain features alone were not sufficient for complete material categorization. Nevertheless, the developed framework establishes a strong foundation for physics-guided drilling condition monitoring and future PHM applications.

6. Future Work

The following research directions are proposed to further enhance the developed framework:

1. - Validate the proposed Health Indicator using actual multilayer drilling specimens containing real material interfaces.
2. - Investigate additional feature domains, including frequency-domain, time-frequency, and wavelet-based features, to improve material categorization and transition detection.
3. - Develop machine learning and deep learning models using the proposed Health Indicator and extracted features for automated material state identification and transition monitoring.
4. - Perform a comparative analysis of different machine learning and deep learning algorithms to identify the most accurate and computationally efficient model.
5. - Apply Explainable Artificial Intelligence (XAI) techniques, including SHAP, LIME, Grad-CAM, and Integrated Gradients, to interpret model predictions and compare their effectiveness in explaining material state decisions.
6. - Incorporate Monte Carlo Simulation to evaluate model robustness and prediction uncertainty under varying operating conditions, sensor noise, and data variability.
7. - Develop a real-time intelligent drilling monitoring framework for industrial PHM applications by integrating the proposed Health Indicator with data-driven decision-making techniques.

7. Acknowledgement

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